



# Manufacturing Report

Industrial sector 2022





# Introduction

We are proud to present you with the fifth edition of the Salary Report for the manufacturing sector. Inside, you will find data gathered from in-depth analysis by experts from Grafton Recruitment and Gi Group over Q1 2022. It encompasses recent factors such as the war in Ukraine or rising inflation in Poland and internationally. However, we do understand that the outcomes of economic, political and social processes are complex enough to most likely leave lasting consequences.

This year's edition of the Report includes data on salaries for almost 140 positions in the following industries: automotive and machining (home appliances, industrial automation), chemistry and FMGC (manufacturing of food, packaging, medical and chemical industries), research and development, engineering centers and a new inclusion of distribution centers. We have been tracking the growth of this sector for a number of years and decided it became a necessary complement to a larger look on the manufacturing sector. Support positions, that is all employees not directly related to manufacturing, quality or logistics, form another important group. A comprehensive selection of positions covered by our report reflects the employment structures in a manufacturing facility.

This report will also provide you with numerous analyses of the manufacturing sector, including overviews of regional educational environments, growth dynamics measured in offers published and the investment potential of each region. An undisputed advantage of this edition is the selection of expert opinions provided by our Customers and Partners of the report. They provide a reference point for particularly current issues, such as: legalization of stay of foreigners, remuneration structure for blue / black collar employees, growth dynamics of the manufacturing industry and the trends and challenges faced by it in 2022.

Since four years, the manufacturing industry is continually being subjected to changes. Past years saw introduction of e-mobility (and the ensuing „recalibration” of the automotive industry), automation and robotization – so-called Industry 4.0, the pandemic or the global supply chain disruption crisis. In 2022 these processes were joined by an armed conflict in Europe and rapid inflation. Despite this complexity, Poland's industrial base retains a good growth dynamic. While in March and April 2022 one could observe PMI decrease for this sector, it seems that this speeding train, once put in motion, will have quite a long braking distance. We hope that our report will provide a valuable commentary on current circumstances. We hope you will have an inspiring read!

***Danuta Protasewicz***  
*Regional Manager Grafton Recruitment*

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## Expert perspective on the manufacturing sector in 2022





# Introduction to the analysis

Michał Kramek  
Senior Data Analyst

According to GUS data, in March of this year, unemployment in Poland has reached 5.4%. From the perspective of number of offers posted on recruitment portals, despite an expectation of refugees from Ukraine saturating the labor market, for several weeks we have been observing first signs of a slowdown. This indicates that despite an increased number of specialists, the geopolitical situation began negatively impacting the demand on the labor market.

In March, in the business sector, the average gross salary amounted to PLN 6,665, representing a growth of 12.4% as compared to the previous year. Similarly, 2021 saw a growth of 8% and 2020 – of 6%. This indicates a consistent trend, aligned with the growing inflation. The salary increase expected by NBP indicates a prediction of approx. 10% y/y\*.

The data obtained during the National Census indicates a population of 38.04 mn (a decrease of 1.2% as compared to 2011). Meanwhile, according to a report by Unia Metropolii Polskich association, the current population amounts to approx. 41.45 mn\*\* (which would indicate a temporary population increase of 7.6% as compared to 2011). At the same time, one can expect that the number of people who decide to stay in Poland after the war in Ukraine is over will likely make up for the loss of population shown by the census data.

\*NBP inflation report [https://www.nbp.pl/polityka\\_pieniezna/dokumenty/raport\\_o\\_inflacji/raport\\_marzec\\_2022.pdf](https://www.nbp.pl/polityka_pieniezna/dokumenty/raport_o_inflacji/raport_marzec_2022.pdf) - pg. 72.

\*\* Urban Hospitality: large growth, challenges and opportunities – Centrum Analiz i Badań, Unia Metropolii Polskich.

5.4%

unemployment rate as of March 2022

PLN 6665

average salary in the business sector  
as of March 2022

9.3–12.2%

expected inflation in 2022 according  
to the March report of the National Bank

38.04 mn

population according to the 2021  
national census

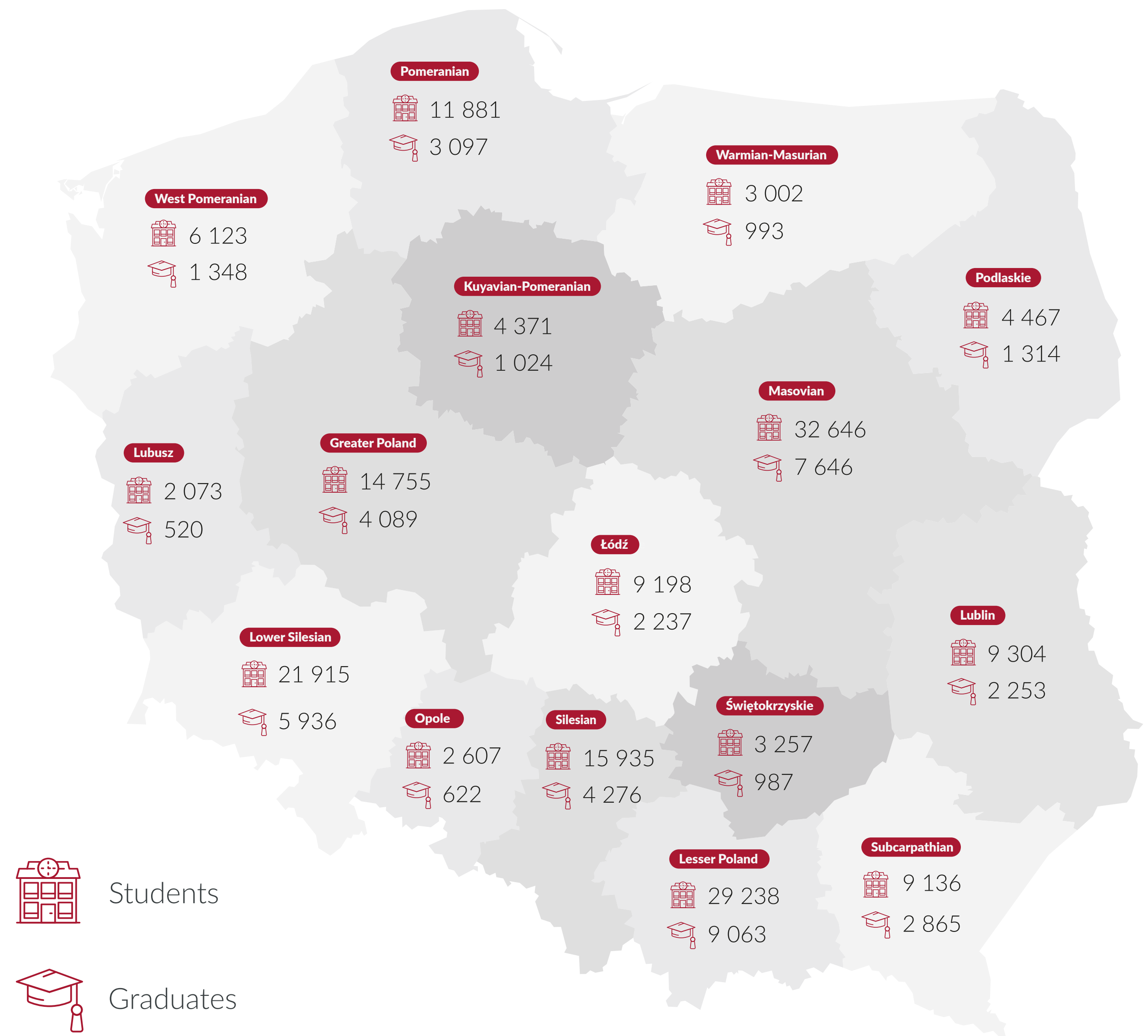
41.45 mn

population according to the  
April 2022 report by Unia  
Metropolii Polskich as of April 2022

# Educational Potential

In the recent years we can see increased difficulties in recruiting employees with specialized technical skills. To appraise the availability of qualified candidates, we have researched the change dynamics of the number of students and graduates of ISCED-F 2013 faculties - technology, industry, construction.

Analysing the y/y change dynamics of the number of students of technology, manufacturing and construction faculties, we have noted a decrease of 3.6% on the national scale, with the overall number of students increasing by approx. 1%. The provinces with most students of technology, manufacturing and construction faculties: Masovia (32,600), Lesser Poland (29,200) and Lower Silesia (21,900). The largest decrease was observed in Świętokrzyskie, Lubusz and Greater Poland provinces.



\* Based on GUS data

# Regional potential

Despite the influx of people from Ukraine making existing data on Poland's demographic somewhat obsolete (approx. 2.5 mn people have crossed the border\*), as of writing GUS is gradually publishing the results of the national census. To assess the current condition of the market, this makes for a good opportunity to capture a static „starting point”. However, one has to account for the movements borne of the current geopolitical situation.

In our analysis, we have split the provinces into three groups, based on their population:

## Group 1 over 2.3 mn

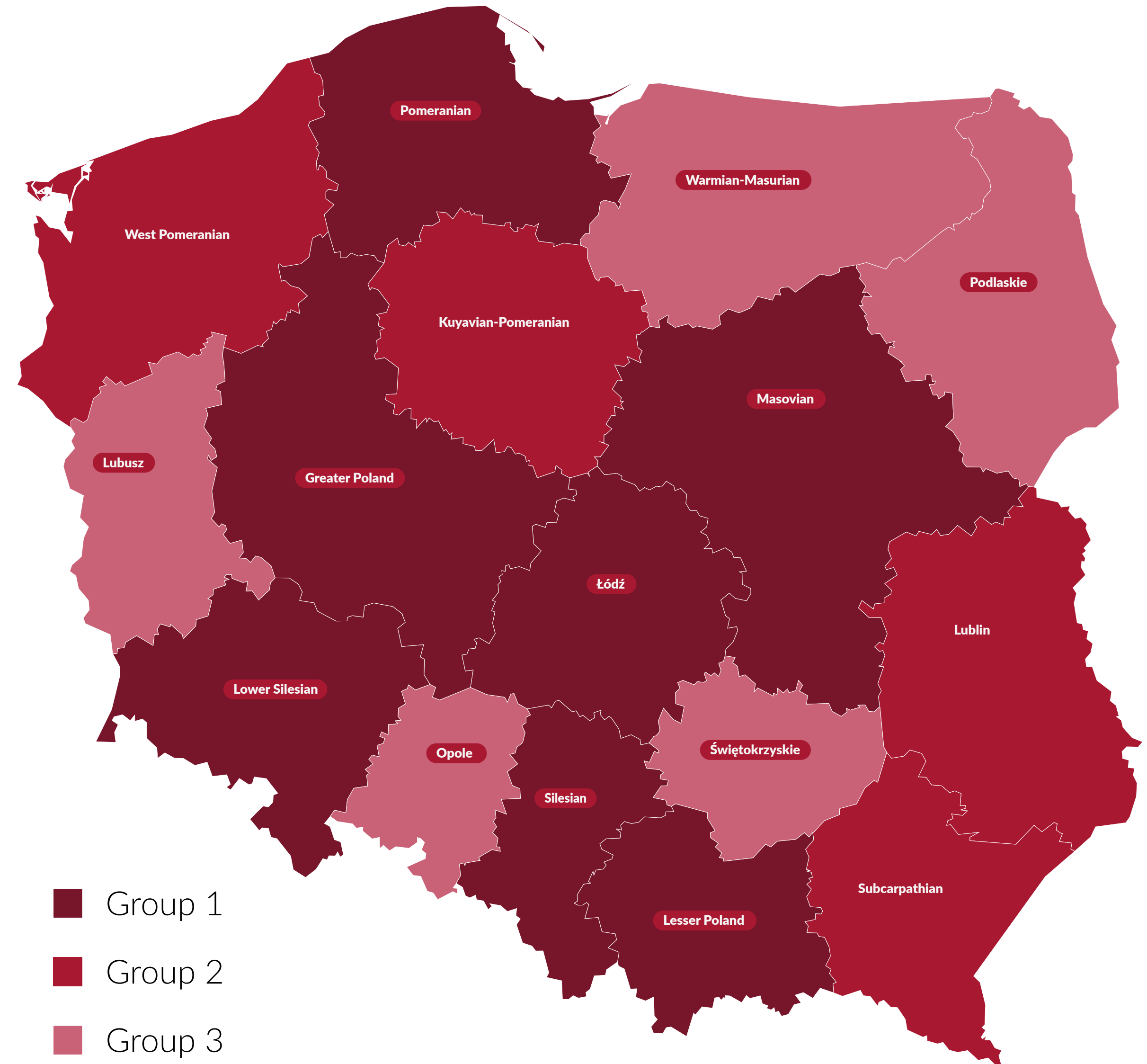
Masovia, Silesia, Greater Poland, Lesser Poland, Lower Silesia, Łódź, Pomerania.

## Group 3 below 1.4 mn

Warmia-Masuria, Świętokrzyskie, Podlaskie, Lubusz, Opole.

## Group 2 over 1.4 mn

Subcarpathia, Lublin, Kuyavia-Pomerania, Subcarpathia, West Pomerania.



- Group 1
- Group 2
- Group 3

\* Urban Hospitality: large growth, challenges and opportunities – Centrum Analiz i Badań, Unia Metropolii Polskich





# Group 1

Group 1 provinces are characterized by low unemployment - 4.6% on average - with the lowest unemployment present in Greater Poland and Silesia. In the researched group, Łódź and Greater Poland provinces have had lower average salaries.

According to the 2021 National Census data, four of the provinces have seen a population growth as compared to 2011. These regions were the only ones among Poland's 16 provinces to grow in population. In the researched group, the largest growth occurred in Masovia and Pomerania (at 4.7% and 3.6%, respectively).

From the housing perspective, the highest vacancy rate were observed in Lesser Poland, Masovia and Łódź provinces. This leads to a conclusion that these regions are best equipped to fulfill the housing needs for the incoming populace - which is of particular interest due to the influx of Ukrainian refugees.

|                | Registered unemployment rate (March) | Average salary in the private sector (March) | Population (in thousands) | % change over 10 years | Area in km2 | Population per km2 | Total apartments (in thousands) | Vacant apartments (total %) |
|----------------|--------------------------------------|----------------------------------------------|---------------------------|------------------------|-------------|--------------------|---------------------------------|-----------------------------|
| LOWER SILESIA  | 4.8%                                 | 6 711                                        | 2 905                     | -0.4%                  | 19 947      | 146                | 1 261                           | 11.4%                       |
| ŁÓDŹ           | 5.7%                                 | 5 986                                        | 2 410                     | -5.1%                  | 18 219      | 132                | 1 060                           | 12.8%                       |
| LESSER POLAND  | 4.5%                                 | 7 063                                        | 3 432                     | 2.8%                   | 15 183      | 226                | 1 297                           | 14.8%                       |
| MASOVIA        | 4.6%                                 | 7 947                                        | 5 515                     | 4.7%                   | 35 558      | 155                | 2 443                           | 13.7%                       |
| POMERANIA      | 5.0%                                 | 7 067                                        | 2 357                     | 3.6%                   | 18 310      | 129                | 937                             | 12.3%                       |
| SILESIA        | 4.2%                                 | 6 677                                        | 4 403                     | -4.9%                  | 12 333      | 357                | 1 815                           | 10.4%                       |
| GREATER POLAND | 3.1%                                 | 5 997                                        | 3 505                     | 1.7%                   | 29 826      | 118                | 1 326                           | 10.8%                       |
| POLAND         | 5.4%                                 | 6 665                                        | 38 036                    | -1.2%                  | 312 679     | 122                | 15 340                          | 12.1%                       |





# Group 2

Group 2 provinces are notable for the highest unemployment rate from among the researched regions - at 7.5% on average. At the same time, when compared to group 1 provinces, their average offered salary in the business sector is lower by approx. PLN 1,000 (at PLN 5,672).

The researched provinces have lower population density than Poland's national average. Simultaneously, got the past 10 years they have seen an above-average population decrease. The largest population drain has been noted in the Lublin province.

From the housing perspective, the highest vacancy rate was seen in Lubusz and West Pomerania provinces. In this case, this is related to outflow of population over the last decade, as show by the results of National Census.

|                   | Registered unemployment rate (March) | Average salary in the private sector (March) | Population (in thousands) | % change over 10 years | Area in km2 | Population per km2 | Total apartments (in thousands) | Vacant apartments (total %) |
|-------------------|--------------------------------------|----------------------------------------------|---------------------------|------------------------|-------------|--------------------|---------------------------------|-----------------------------|
| KUYAVIA-POMERANIA | 7.6%                                 | 5 881                                        | 2 027                     | -3.4%                  | 17 972      | 113                | 791                             | 9.7%                        |
| LUBLIN            | 7.2%                                 | 5 345                                        | 2 052                     | -5.7%                  | 25 122      | 82                 | 798                             | 13.5%                       |
| SUBCARPATHIA      | 8.1%                                 | 5 432                                        | 2 093                     | -1.6%                  | 17 846      | 117                | 704                             | 11.1%                       |
| WEST POMERANIA    | 7.1%                                 | 6 030                                        | 1 658                     | -3.8%                  | 22 892      | 72                 | 692                             | 11.8%                       |
| POLAND            | 5.4%                                 | 6 665                                        | 38 036                    | -1.2%                  | 312 679     | 122                | 15 340                          | 12.1%                       |





# Group 3

Group 3 provinces are characterized by an average unemployment rate of 6.7% and average salary in the business sector amounting to PLN 5,583. This means that despite a lower unemployment (as compared to group 2), the salaries are also lower on average.

This group has the lowest population density among the researched regions, with Warmia-Masuria and Podlaskie provinces having the least inhabitants per square kilometer. At the same time, it bears mentioning that according to the national census data, all provinces in Poland have seen a population decrease. Having said that, group 3 provinces have had the largest average outflow of inhabitants, with Świętokrzyskie and Opole provinces being particularly impacted by this.

From the housing perspective, the largest vacancy rate was noted in Podlaskie province, the second highest result following closely after that of Lesser Poland.

|                | Registered unemployment rate (March) | Average salary in the private sector (March) | Population (in thousands) | % change over 10 years | Area in km2 | Population per km2 | Total apartments (in thousands) | Vacant apartments (total %) |
|----------------|--------------------------------------|----------------------------------------------|---------------------------|------------------------|-------------|--------------------|---------------------------------|-----------------------------|
| LUBUSZ         | 4.8%                                 | 5 616                                        | 991                       | -3.1%                  | 13 988      | 71                 | 394                             | 9.6%                        |
| OPOLE          | 6.0%                                 | 6 108                                        | 954                       | -6.1%                  | 9 412       | 101                | 361                             | 8.9%                        |
| PODLASKIE      | 7.0%                                 | 5 519                                        | 1 154                     | -4.0%                  | 20 187      | 57                 | 465                             | 14.7%                       |
| ŚWIĘTOKRZYSKIE | 7.3%                                 | 5 354                                        | 1 197                     | -6.6%                  | 11 711      | 102                | 465                             | 13.5%                       |
| WARMIA-MASURIA | 8.6%                                 | 5 316                                        | 1 382                     | -4.8%                  | 24 173      | 57                 | 532                             | 10.2%                       |
| POLAND         | 5.4%                                 | 6 665                                        | 38 036                    | -1.2%                  | 312 679     | 122                | 15 340                          | 12.1%                       |





## Investment potential

According to observations of Grafton's experts, investors in manufacturing, logistics and R&D industries are primarily choosing cities in group 1 provinces located in the southern parts of the country. Despite a low unemployment rate and relatively high competition, these regions are Poland's leaders in terms of both the number of graduates of technology, manufacturing and construction faculties and overall population density. From an investor viewpoint, Poland's southern regions are considered on par with Romania, Czechia and Slovakia as investment destinations, due to factors such as high number of specialists, as well as students and graduates of technology, manufacturing and construction faculties.

Due to the current geopolitical situation, directly impacting all areas of life, the markets see increased investor activity, as business development strategies shift. Thus, due to lack of stability, some hitherto planned investments in Ukraine or Belarus are either being abandoned or relocated to Poland.

*Michał Kramek*  
*Senior Data Analyst*





Polska  
Strefa Inwestycji



POMORSKA  
SPECJALNA STREFA  
EKONOMICZNA

# Przemysław Sztandera

President of the Board

**// Thanks to the Polish Investment Zone, entrepreneurs from the entire country can apply for income tax exemption over a period of 12 to 15 years.**

From the perspective of Pomeranian Special Economic Zone, the machining, metal, packaging, furniture and chemical industries were the main beneficiaries of the Polish Investment Zone. The area managed by PSEZ, namely Kuyavia-Pomerania and the eastern part of Pomeranian province, has seen a notable upturn in investments by small and medium-sized enterprises. Companies from these sectors know that each investment in development is a guarantee of retaining market competitiveness and obtaining new clients. The Economic Zone strives to utilize the local potential, as oftentimes family companies bring innovative ideas and, thanks to tax exemptions and investment counseling, are able to invest in their own development and become a serious market player.

Polish Investment Zone allows entrepreneurs from the entire country can apply for income tax exemptions for a period of 12 to 15 years. This solution is offered to companies planning new investments, regardless of their size. Receivable support depends on either the investment size or employment costs, the location and status of a given company. Despite a noticeable trend of increasing SME investments, many of them still reflexively associate the Polish Investment Zone with only large companies. This is a huge mistake, as it is small, medium and microenterprises who receive most support.



The maximum level of support offered in Pomerania is 30% for large enterprises, 40% for medium-sized companies and 50% for small and micro-enterprises. In Kuyavia-Pomerania they can receive up to 40%, 50% or 60%, respectively, of their qualified costs. What does it mean in practice? For example, a medium-sized business in Pomerania planning to invest PLN 10 mn into expanding their production facility could receive PLN 4 mn in income tax exemption. The company will be able to skip this tax up until it reaches that amount or the support decision expires. In case of an analogous company in Kuyavia-Pomerania, the tax exemption would amount to PLN 5 mn.

It bears mentioning that the time between submission of a set of simple documents and receiving the decision does not exceed 14 days. Furthermore, we support the investors in preparing their application.

To receive a positive decision, a company needs to fulfill quantitative criteria - a minimum investment expenditure, dependent on its size and the unemployment rate in a given poviát. The company also needs to fulfil qualitative criteria in two categories. The first of these promotes activities related to economic growth, such as scientific development, membership in Key National Cluster or the status of a small, medium or micro-enterprise. The second category concerns social development, through means such as offering stable employment, maintaining low environmental impact or supporting employee development through improving their qualifications and co-operating with vocational schools.

Polish Investment Zone is also important from a regional viewpoint. It banks on creating stable, well-paid positions, education and developing human resources. This is one of many reasons Pomeranian Special Economic Zone helped create the Industrial Robot Programming Center, allowing to both increase employee competences and train vocational school students. In effect, companies can obtain qualified personnel and employees can receive agreeable employment terms.





## Artur Małkiewicz

Recruitment Manager

**// Over 3 milion Ukrainians have arrived in Poland to flee war. At the same time, almost a million people have returned from Poland to Ukraine, including almost 200,000 males.**

According to estimates, Poland's labor market will be enlarged by approx. six hundred thousand employees from Ukraine, leading to a record low unemployment. However, this leads to a question of whether this will solve the shortage of „working hands” currently faced by employers in the manufacturing industry?

Two months have passed since Russia's military attack on Ukraine. In this time, over 3 mn Ukrainian refugees reached Poland, fleeing the war. At the same time, almost a million people have returned from Poland to Ukraine, including almost 200,000 males. Such an influx of people might have a large impact on employee management policies of Polish manufacturing businesses. Some of them are very optimistic, seeing a potential for much easier recruitments. However one should remember the situation is highly dynamic. According to the statistics, almost half of refugees arriving in Poland are children and many of the adults are either elderly or women with no access to systemic means of caring for the children, preventing them from seeking stable employment.

Shortage of workers on essentially male-only positions is a key challenge faced by the manufacturing industry. This is further compounded by the often difficult language barrier. Of course, some businesses



are able to employ people who do not speak Polish, but even them can be overwhelmed by the sheer number of such employees hired in a single period.

The entrepreneurs should therefore prepare several scenarios for strategic employee management, taking into the account: availability of appropriate candidates, their capability to train new employees for their roles and the support they can offer to refugees. They should encompass several variants of applicable procedures, reflecting the real needs and capabilities of Ukrainian employees. Let's remember that they not only include people who plan to remain and work in our country, but also encompass those who wish to leave for Western countries or bring in their families from Ukraine.

There are more challenges related to employing Ukrainians, but thankfully entrepreneurs are being offered a number of legislative improvements, minimizing formalities required during the recruitment process - for example lifting of the need to legalize employee's stay under the condition of a legal stay within 18 months from the outbreak of the war. Employees from Ukraine can now change their employment easier, because residence cards are given out without the need to assign them to a specific employer, therefore removing the formal obstacle to changing one's employment. Of course, this also imparts a need to diversify the sources and profiles of employees, as nowadays candidates from Ukraine can change workplaces just as easily as Poles.

The coming months will prove very dynamic and somewhat unpredictable for the manufacturing industry, as migration of Ukrainian employees will greatly impact both candidate availability and the continuity of production. There are many proposals for further governmental regulations, both in Poland and Ukraine. They could either lower or enlarge the availability of Ukrainian employees - and primarily male at that. To successfully implement one's business goals in manufacturing, it is key not only to plan several variants for support, but also quickly adapt to the changing labor market.





# Barbara Turska

BT International

## **Legalizing foreigners' stay in 2022: new challenges, new opportunities**

For the past several years, we have seen an increased number of foreigners coming to Poland. We more and more of them where we work and live. At the same time, negative demographic processes (such as low natural growth, aging of society) have been causing a shortage of Polish workforce. It bears mentioning that a large number of Poles, especially those in working age, lives abroad. All of these factors causes employers, who face increased difficulties in filling vacancies, to turn towards hiring foreign employees. Currently this primarily concerns immigrants from Ukraine and Belarus, but also includes people from farther corners of the world, such as India, Bangladesh, Indonesia or Nepal. Current forecasts suggest this trend will last.

The sudden increase in the number of foreigners taking up work in Poland brings a number of challenges on both the national and local levels. That is why in the recent months a large number of legal changes concerning employment of foreigners has been introduced. The goal of these regulations is to facilitate flexible registration of work orders concerning foreigners and optimizing procedures related to granting them residence permits, especially in case of temporary residence and work permits. The new regulations are also meant to improve other, pre-existing solutions concerning foreigners, such as those related to hiring of employees transferred within a given company.





## What do positive changes mean exactly?

The new laws extend the list of circumstances not requiring changing or issuing new temporary residence and work permits. This makes procedures necessary to obtain these permits shorter and simpler.

However, a law introduced on 12.03.2022 regulates matters related to the war in Ukraine. Its key provisions concern the employment and residence of refugees. It also specifies additional, temporary rules for legal stay of Ukraine's citizens on the territory of Poland:

- If a citizen of Ukraine has legally entered Poland between February 24 2022 and the day specified in an implementing regulation of the Council of Ministers and declares willingness to remain on its territory, their stay is deemed legal for 18 months, starting from February 24 2022;
- A child's stay is also deemed legal if it was born on Poland's territory while its mother was subject to the previous clause;
- If a citizen of Ukraine holding a Pole's Card has left Ukraine after February 24 2022 and then arrived on the territory of Poland before the day specified in an implementing regulation of the Council of Ministers and declares willingness to remain on its territory, their stay is deemed legal for 18 months, starting from February 24 2022;

The laws concerning legalization of stay are not applied to the citizens of Ukraine already having a permanent residence permit, temporary stay permit or a long-term EU resident status. They also don't concern people with a refugee status, subsidiary protection, permit for tolerated stay or an authorization to stay for humanitarian reasons who also:

a) submitted applications to be granted international protection on the territory of Poland or have had such submissions submitted in their name,

b) declared willingness to submit applications to be granted international protection on the territory of Poland based on Art. 28(1) or Art. 61(1) of the June 13 2003 Act on Providing Protection to Foreigners on the Territory of Republic of Poland (Journal of Laws of 2021, items 1108 and 1918 and of 2022, item 583) or who are subjects of such declarations.

There are far more legal changes concerning foreigners, making it very worthwhile to learn of them in-depth. Furthermore, both employers and foreigners can make use of special information points at the voivodeship offices or utilize support of specialized companies, which are present in each province.





# Sector growth dynamic and regional potential comparison

To compare indicators and basic facts of the Manufacturing sector, we have chosen a recruitment portal where over 20% of job offers concern physical labor (provincial average amounts to 24.3%). We wanted to make sure that it is well-connected with the kind of candidates we wish to research.

According to our analysis, the average ratio of offers per 1,000 inhabitants amounts to 0.67, with Opole, West Pomerania and Warmia-Masuria provinces having the highest demand (at 0.87 for the former two and 0.81 for the latter). Podkarpackie, In turn, Lesser Poland and Podlaskie provinces have the lowest saturation (at 0.48, 0.54 and 0.54 again, respectively).

|                   | Registered unemployment rate (March) | Population (in thousands) | Offers per 1,000 inhabitants | % physical labor | % engineering, servicing, technology, installation | % Logistics / Transport / Shipping | % Servicing / Technology / Assembly / Engineering / Design |
|-------------------|--------------------------------------|---------------------------|------------------------------|------------------|----------------------------------------------------|------------------------------------|------------------------------------------------------------|
| Lower Silesia     | 4.8%                                 | 2904.9                    | 0.71                         | 26.2%            | 19.2%                                              | 8.4%                               | 19.2%                                                      |
| Kuyavia-Pomerania | 7.6%                                 | 2027.3                    | 0.57                         | 23.9%            | 17.9%                                              | 3.5%                               | 10.0%                                                      |
| Lublin            | 7.2%                                 | 2052.3                    | 0.55                         | 22.2%            | 16.6%                                              | 3.5%                               | 9.0%                                                       |
| Lubusz            | 4.8%                                 | 991.2                     | 0.79                         | 28.4%            | 23.3%                                              | 2.9%                               | 8.8%                                                       |
| Łódź              | 5.7%                                 | 2410.3                    | 0.68                         | 28.2%            | 16.9%                                              | 6.9%                               | 13.4%                                                      |
| Lesser Poland     | 4.5%                                 | 3432.3                    | 0.54                         | 18.5%            | 15.4%                                              | 5.0%                               | 13.7%                                                      |
| Mazowieckie       | 4.6%                                 | 5514.7                    | 0.72                         | 20.3%            | 15.0%                                              | 14.2%                              | 28.9%                                                      |
| Opole             | 6.0%                                 | 954.1                     | 0.87                         | 25.6%            | 18.1%                                              | 2.7%                               | 7.3%                                                       |
| Subcarpathia      | 8.1%                                 | 2093.3                    | 0.48                         | 23.9%            | 20.8%                                              | 3.5%                               | 10.1%                                                      |
| Podlaskie         | 7.0%                                 | 1154.3                    | 0.54                         | 26.3%            | 25.3%                                              | 1.2%                               | 7.7%                                                       |
| Pomerania         | 5.0%                                 | 2357.3                    | 0.73                         | 21.5%            | 20.2%                                              | 5.5%                               | 16.7%                                                      |
| Silesia           | 4.2%                                 | 4403.0                    | 0.58                         | 24.3%            | 15.1%                                              | 9.9%                               | 18.5%                                                      |
| Świętokrzyskie    | 7.3%                                 | 1196.6                    | 0.71                         | 22.3%            | 19.1%                                              | 2.5%                               | 7.8%                                                       |
| Warmia-Masuria    | 8.6%                                 | 1382.2                    | 0.81                         | 26.6%            | 18.8%                                              | 3.1%                               | 10.3%                                                      |
| Greater Poland    | 3.1%                                 | 3504.6                    | 0.56                         | 25.3%            | 16.1%                                              | 8.0%                               | 15.3%                                                      |
| West Pomerania    | 7.1%                                 | 1657.7                    | 0.87                         | 25.8%            | 17.8%                                              | 5.2%                               | 12.3%                                                      |



As for the structure of job offers, the Lubusz, Łódź and Warmia-Masuria provinces have the largest share of physical jobs among the offers provide. This means that a relatively large share of vocational offers concerns physical work. This indicator should be compared to the unemployment rate and local population – one can notice that Warmia-Masuria and Podlaskie provinced (respectively thrid and fourth in terms of this metric) have an unemployment rate in excess of 7%, despite a similar population and above-average (in the national scale) demand for physical work. We believe this is due to lower levels of professional activation in these regions. In case of the engineering-related group, Podlaskie, Lubusz and Podkarpackie provinces lead in terms of the demand structure.

In terms of sector's maturity and population, Masovia, Lower Silesia, Silesia, Greater Poland and Lesser Poland are leading. The former two have a large demand, indicating high dynamics of the manufacturing sector. Silesia and Greater Poland, the traditional bastions of the automotive industry, which for the past few years has been undergoing difficulties, and Lesser Poland are seeing relatively small interest. However, it seems that given their human resources and know-how these regions retain an investment potential. Due to the nature of positions offered, physical jobs not requiring specific qualifications appear to be most popular, followed by engineering, servicing, technical and assembly positoins. Logistics, transport and shipping jobs are third in terms of demand analysis. In terms of demand allocation, Podlaskie and Pomerania provinces are particularly interesting, due to their relatively uniform demand for each of the categories.





# Warehouse Space

## CBRE

### Beata Hryniewska

Director of Warehouse and Industrial  
Space Department at CBRE

**// Nothing will stop the warehousing sector.  
The demand will stay high, despite slowing supply.**

Over the past five years, warehouse space in Poland has doubled. The sector gained momentum during the pandemic, leading to a series of record years. 3.08 mn m<sup>2</sup> of warehouse space has entered the market, with the demand reaching a historical high of 7.53 mn m<sup>2</sup>. This period has also brought a record amount of space in development and a multi-year low vacancy rate. In 2021, the warehousing market accounted for over a half of all commercial real estate investments. 2022 will also be a year of warehouses, despite the current situation beyond our eastern border likely also having an impact on it.

The warehousing sector keeps growing, despite difficult external circumstances. The pandemic did not slow its momentum and the war in Ukraine is also unlikely to do so. Having said that, the war has highlighted several global issues, including the collapse of supply chains and limited availability of resources and goods. The consequences of these circumstances will be keenly felt in the Central-Eastern Europe, including Poland. They will impact supply, demand and upcoming investments alike.



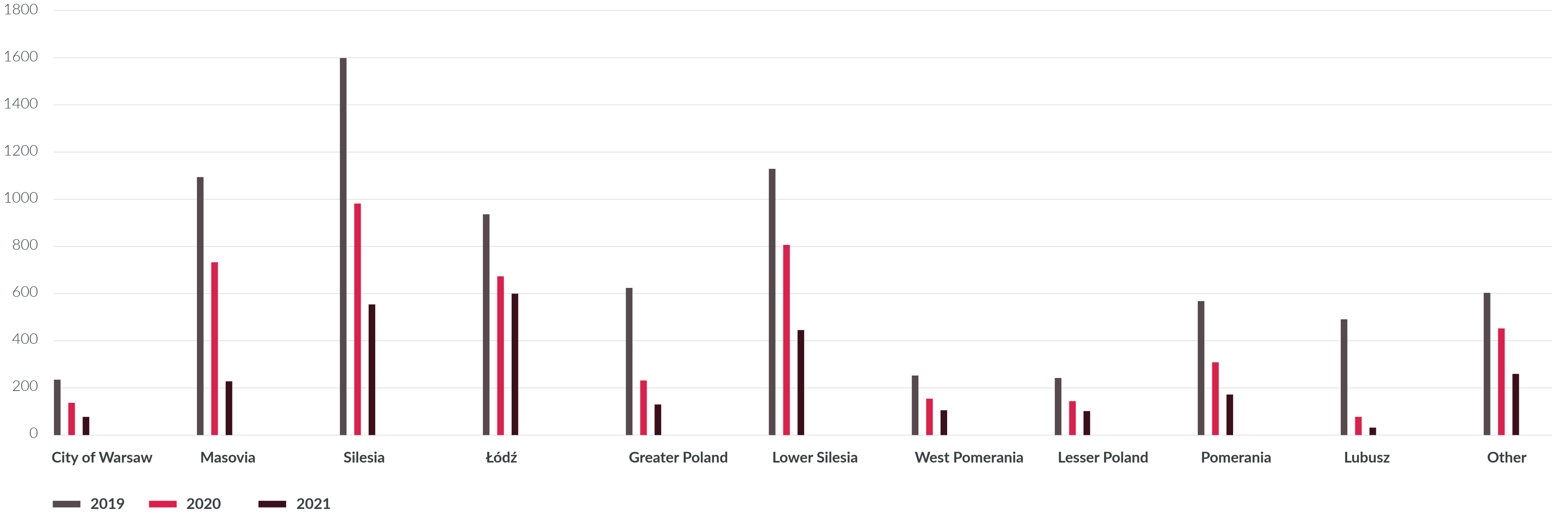


Higher costs, lower supply

In 2021, the costs of warehouse construction have risen by over 20 percent. This year will make this situation even worse, due to a limited availability of resources imported from Russia and Ukraine. This extends the construction time of a warehouse, especially given the fact the shortage of resources is paired with an outflow of workers from Ukraine. According to estimates, its scale might have reached even 0.5 mn people. While construction projects which began development before the war will likely

continue (due to having already secured building materials), but launching new developments might prove difficult. This poses a risk of supply gap, especially in light of historically low vacancy rate by the end of 2021. Therefore, limited construction plans combined with continuously high demand will likely cause a supply shortage.

New space put into regional markets over the past three years



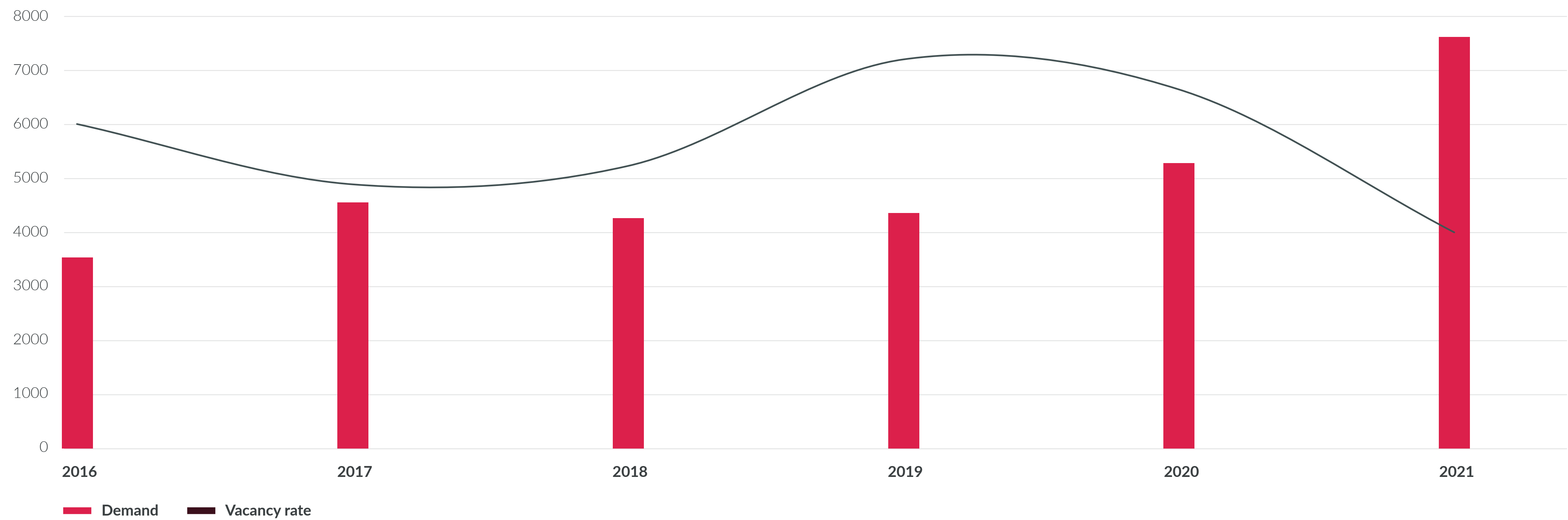




The increased construction costs are paired with rising inflation, which also includes energy prices. This will only strengthen the trend that was already bound to increase in intensity: an increased focus on energy efficient buildings. We expect green construction to gain increased importance in the warehousing sector. We see an environmental focus on the side of both tenants and developers. The former are increasingly often taking ESG rating into the account when choosing their location.

The latter count on sustainable solutions both increasing the attractiveness of a building and distinguishing them from among other developers. Furthermore, a warehouse is a long-term investment, which must remain attractive for investors in a handful of years. Today, no-one doubts that sustainable construction is the way of the future: first, because the availability and variety of green solutions is increasing and second, because numerous directives and regulations demand use of such green solutions.

**Demand and vacancy rate**





### **The demand will not fall...**

One should expect that the demand for warehouses will remain high during this year. This will be caused by, among other factors, companies relocating from Ukraine, increased consumption generated by approx. 2 mn refugees and the continuously strong dynamics of the e-commerce sector. One should remember that the demand on the warehousing market is generated primarily by e-commerce companies. In Poland, online sales are growing, with clients turning to it during the pandemic retaining their newfound habits, in turn stimulating the growth of infrastructure necessary for efficient handling of an increasing number of orders. This includes, for example, distribution warehouses or return centers. Due to the fact many companies utilize services of specialized logistics companies, further demand for warehousing space is created. It's possible, of course, that the economic uncertainty will cause consumer trust to fall, which combines with the inflation could lead to lowered buying power and a number of companies, primarily in the retail sector, to put their expansion plans on hold. However, it will definitely not stop the development of the sector at large. The new wave of COVID-19 in China will likely lead to subsequent disruption of production and logistics, causing delays in the arrival of ordered goods in Europe and Poland. The companies will want to increase their warehoused stocks to protect against further supply chain disruptions. In this way, they drive demand for new warehouse space. The negative impact of current circumstances on the warehousing industry will be felt most keenly by the automotive industry. It is currently facing component shortage difficulties and considering relocation of facilities rendered inoperable in Ukraine or away from Russia and Belarus.

### **... but the prices and rent will rise**

Developer costs are rising due to construction material prices, in turn impacting the rents, which have remained stable up until Q3 of the previous year. Afterwards, this has changed. Contractual rents have increased by 5-15%, depending on market, and will most probably continue to rise. We will take a closer look at select locations. In Warsaw, rent rates have increased by 6%, in Cracow by 5% and

by 4% in Łódź and Katowice. It bears mentioning that rent rates in Poland have remained below the Western and Central-Eastern Europe average for a long time. Therefore, they can remain competitive despite the increase. However, the PLN-EUR exchange rate is not stable, and the common practice of invoicing rent in EUR and paying them in PLN might deter tenants.

### **Investment Market**

Industrial and Logistics assets are perceived as a strong investment product, a fact proved throughout the pandemic. In 2021, the total value of investments has reached EUR 5.71 mn, encompassing a record high of 164 transactions. Warehousing market has accounted for over a half of this sum. Most investors came from various European countries, with Polish capital only accounting for 4%. Importantly, in the last handful of years we have not seen any notable capital transfers from Russia, Ukraine and Belarus on the industrial and warehousing real estate markets. This indicates that a loss of these investment will not a strong impact on the overall results. The investors will remain interested in warehouses, even despite a temporary cautiousness, as they wait to see the impact of the still-unfolding war in Ukraine.





## Salary report

Research conducted by experts of Grafton Recruitment Sp. z o.o. and Gi Group Sp. z o.o. in Q1 2022. Gross monthly salaries expressed in PLN as minimum-maximum ranges.

Blue and black collar salaries encompass both the base pay and an average bonus. The research has been conducted on the basis of analysing 9,567 current candidate records and salary budgets of 189 manufacturing companies. Specialist positions assume at least 3 years of occupational experience. Management and leadership positions assume a minimum of a year as a manager or leader.







# Tomasz Tarabuła

Regional Director

## Salary structure, or how to attract a production employee

For two years, the labor market has been undergoing continuous changes. In this time, we have seen a labor crisis caused by the COVID-19 epidemic, difficulties with component availability, supply chain disruptions, war in Ukraine and runaway inflation. All these problems have forced many entrepreneurs to reduce personnel and the periods of cost saving mixed with periods of increased demand for employees. The automotive industry has been the most heavily impacted by this situation.

All these difficulties require the companies to appropriately manage the costs of their activities, constituting in large part of employee salaries. This oftentimes touches on bonuses complementing the base salary. From employer's viewpoint, the larger share bonuses take in employee's overall pay, the easier it is to manage the costs and adjust them to company's current situation.

However, what about the employee's perspective? While they feel impact of the crises just as strongly, temporary job reductions are always followed by a recovery and a return to employee's market - a situation in which it is them who choose between many offers. Most of the time, a candidate will choose the best paid job - it's fairly obvious that a higher base salary makes it easier to attract an employee. But what about the bonuses? Which of them are considered most attractive and which are skippable?





Discretionary bonus – is an effective way of deterring potential candidates. Even providing a range does not help: candidates openly admit that they consider the discretionary bonus to be highly uncertain, as it is oftentimes unrelated to their own productivity.

Bonus dependent on company results – while not received as negatively as the discretionary bonus, it does little to increase interest in the offer, particularly in case of large businesses.

Bonus dependent on employee's results – when offering such bonus, it's best to clearly and precisely define employer's requirements and their direct impact on employee's pay. The simpler and clearer the remuneration system is, the stronger their engagement will be.

Statutory bonus – a fixed bonus giving the employee a sense of pay stability and retains some flexibility for the employer. The employees tend to say „you have to put in some effort not to get it“. A common form of such a bonus is the frequency premium.

Bonuses related to the work system or harmful work conditions - these bonuses are meant to make offers with these negative traits more attractive. They provide a good solution for employees motivated primarily by money.

The offer can also be made to stand out by proposing a Christmas bonus, holiday bonus, referral bonus, or the increasingly more popular starting bonus, meant to entice a candidate to take the job here and now. It usually takes the form of additional funds payable upon three months of work or paid out over that period.

What kind of bonus is most popular? The one that is best defined, related to employee's performance or of a fixed nature. However, one should remember that ultimately the candidates are more likely to prioritize base salaries offered.

Perhaps it is best to add employee benefits? Which ones would be most successful in attracting candidates?

Regrettably, many of the benefits, such as medical care, sports vouchers or co-financed insurance ceased to be perceived as benefits and became more of a baseline standard expected by employees. Increasingly often, candidates gloss over this aspect of job offers and not consider it impactful on their decision. However, this does not mean one should simply abandon the benefits. It is still worth to invest in the kind that will make our offer stand out.

For example, blue and black collar workers will appreciate access to fairly priced vending machines, co-financing or full reimbursement of workplace canteen meals or certification (such as forklift, crane or welder certification). The latter is highly attractive to those wishing to expand their competences. Such elements are worth investing in and bragging about, especially given that candidates themselves ask about them during job interviews.

When choosing the benefits to offer, one should have the target group in mind. What would attract a white collar candidate will not necessarily be appreciated by blue or black collar workers. However, all these groups will find access to streaming platforms such as TIDAL, Spotify or Netflix attractive. In fact, this benefit is becoming increasingly common, especially in offers aimed at younger candidates. Proper selection of bonuses and benefits might convince a candidate to join your company. Afterwards, it is simply a matter of onboarding and keeping the employee – but that is an entirely different topic.





# Distribution centers







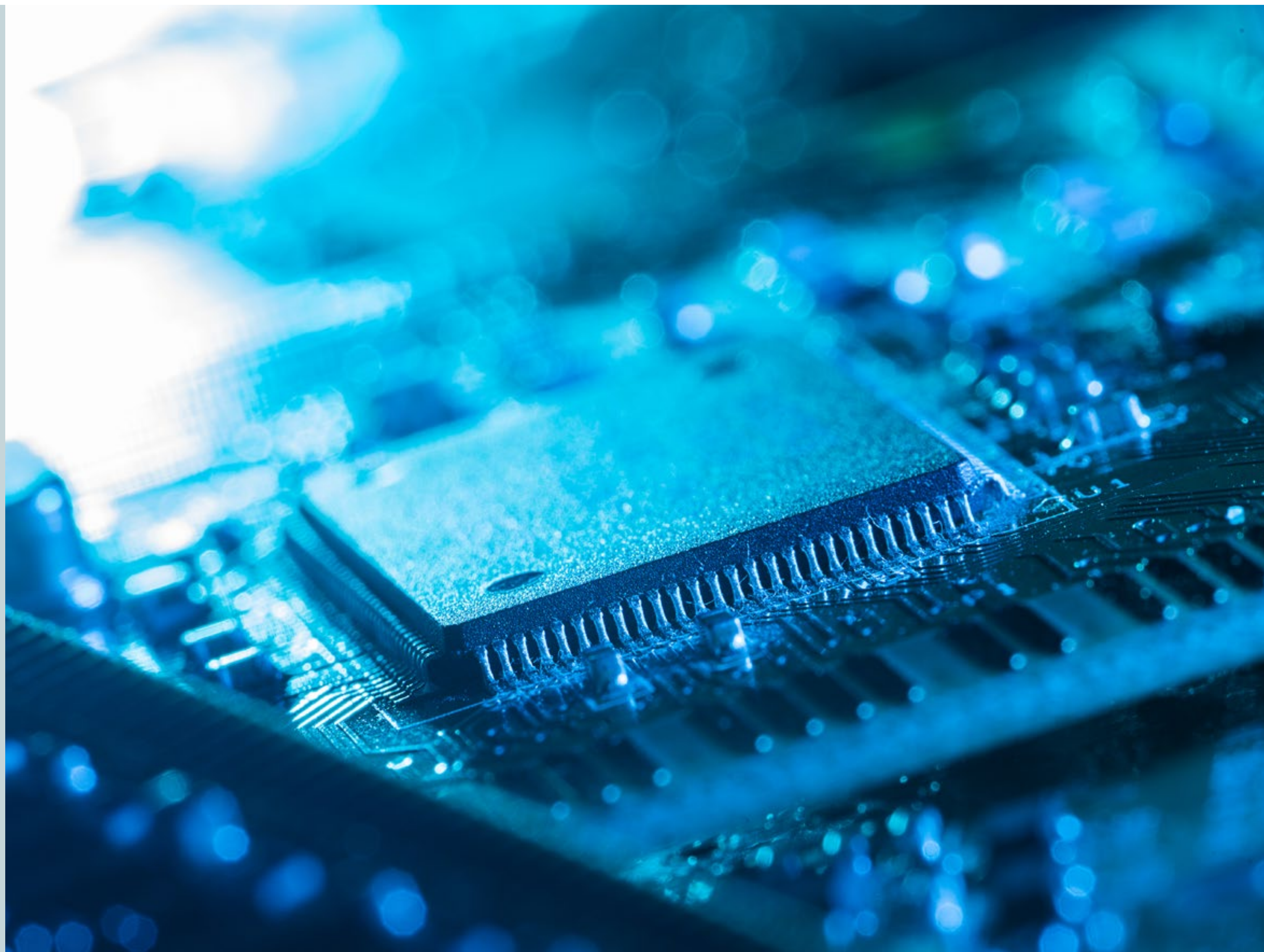
## Distribution centers

|                              | 2022          |        |        |        |               |        |         |        |         |        |                |        |           |        |
|------------------------------|---------------|--------|--------|--------|---------------|--------|---------|--------|---------|--------|----------------|--------|-----------|--------|
|                              | Lower Silesia |        | Łódź   |        | Lesser Poland |        | Masovia |        | Silesia |        | Greater Poland |        | Pomerania |        |
|                              | Lodz          | max    | min    | max    | min           | max    | min     | max    | min     | max    | min            | max    | min       | max    |
| Distribution Center Manager  | 16 000        | 24 000 | 15 000 | 20 000 | 18 000        | 24 000 | 16 000  | 25 000 | 16 000  | 24 000 | 15 000         | 25 000 | 16 000    | 24 000 |
| Supply Chain Manager         | 14 000        | 18 000 | 13 000 | 17 000 | 15 000        | 18 000 | 15 000  | 20 000 | 15 000  | 18 000 | 15 000         | 21 000 | 16 000    | 18 000 |
| Logistics Manager            | 12 000        | 17 000 | 11 000 | 15 000 | 13 000        | 16 000 | 14 000  | 18 000 | 13 000  | 17 000 | 13 000         | 17 000 | 12 000    | 17 000 |
| Logistics Specialist         | 6500          | 9 000  | 6 000  | 8 000  | 6 500         | 8 500  | 7 000   | 9 000  | 6 500   | 9 000  | 6 000          | 8 500  | 6 500     | 9 000  |
| Warehouse Manager            | 9000          | 14 000 | 8 500  | 13 500 | 9 000         | 14 000 | 11 000  | 16 000 | 9 000   | 14 000 | 9 000          | 14 000 | 9 000     | 14 000 |
| Warehouse Worker             | 3300          | 4 500  | 3 200  | 4 200  | 3 300         | 4 500  | 3 800   | 4 800  | 3 300   | 4 500  | 3 400          | 4 500  | 3 300     | 4 500  |
| Quality Controller           | 3500          | 4 800  | 3 500  | 4 500  | 3 500         | 4 800  | 4 000   | 5 000  | 3 500   | 4 800  | 4 000          | 5 000  | 3 500     | 4 800  |
| Forklift Driver              | 4500          | 6 000  | 4 000  | 5 500  | 4 500         | 5 500  | 5 000   | 6 000  | 4 500   | 5 800  | 4 500          | 6 000  | 4 500     | 5 800  |
| Transport Manager            | 9000          | 13 000 | 9 000  | 13 000 | 9 000         | 13 000 | 9 500   | 14 000 | 9 000   | 13 000 | 9 000          | 13 000 | 9 000     | 13 000 |
| Transport Specialist         | 5500          | 7 000  | 5 000  | 6 500  | 5 500         | 7 000  | 5 500   | 7 500  | 5 500   | 7 000  | 5 000          | 7 000  | 5 000     | 7 000  |
| Freight Forwarder            | 5500          | 8 500  | 5 000  | 8 000  | 5 500         | 8 500  | 7 000   | 9 000  | 5 500   | 8 500  | 5 500          | 8 500  | 5 000     | 8 500  |
| Logistics Solutions Designer | 7500          | 9 500  | 7 500  | 9 000  | 7 500         | 9 500  | 7 500   | 10 000 | 7 500   | 9 500  | 7 500          | 10 000 | 7 500     | 9 500  |
| Purchasing Manager           | 10000         | 15 000 | 9 000  | 14 000 | 10 000        | 15 000 | 12 000  | 17 000 | 10 000  | 15 000 | 11 000         | 16 000 | 10 000    | 15 000 |
| Purchasing Specialist        | 7000          | 10 000 | 6 500  | 9 500  | 7 500         | 10 000 | 8 000   | 12 000 | 7 500   | 10 000 | 7 500          | 11 000 | 7 000     | 10 000 |
| Material Planner             | 7000          | 9 500  | 6 500  | 9 000  | 7 000         | 9 500  | 7 500   | 10 000 | 7 000   | 9 500  | 7 000          | 9 500  | 7 000     | 9 000  |
| Customer Service Manager     | 11000         | 14 000 | 9 000  | 13 000 | 11 000        | 14 000 | 11 000  | 16 000 | 11 000  | 14 000 | 11 000         | 15 000 | 11 000    | 14 000 |
| Customer Service Specialist  | 6000          | 8 500  | 4 500  | 7 500  | 6 500         | 8 500  | 4 500   | 5 500  | 6 500   | 8 500  | 5 500          | 7 500  | 6 000     | 8 500  |
| Shift Supervisor             | 5500          | 7 000  | 5 500  | 7 000  | 5 500         | 7 000  | 6 000   | 8 000  | 5 500   | 7 500  | 6 000          | 8 000  | 5 500     | 7 000  |
| Foreman                      | 5000          | 6 000  | 5 000  | 6 000  | 5 000         | 6 000  | 5 500   | 7 000  | 5 000   | 6 000  | 5 500          | 6 500  | 5 000     | 6 000  |
| Facility Technician          | 5500          | 7 000  | 5 500  | 6 500  | 5 500         | 7 000  | 6 000   | 7 000  | 5 500   | 7 500  | 5 500          | 7 500  | 5 000     | 7 000  |
| Facility Manager             | 10000         | 15 000 | 9 500  | 14 000 | 10 000        | 15 000 | 12 000  | 18 000 | 10 000  | 15 000 | 10 000         | 15 000 | 10 000    | 14 000 |





R&D







R&D

|                                 | 2022   |        |
|---------------------------------|--------|--------|
|                                 | min    | max    |
| Hardware                        |        |        |
| R&D Director                    | 28 000 | 48 000 |
| Mechanical Design Manager       | 18 000 | 25 000 |
| Mechanical Design Team Leader   | 15 000 | 22 000 |
| Mechanical Design Engineer      | 8 500  | 14 000 |
| Manager of Hardware Electronic  | 20 000 | 28 000 |
| Hardware Electronic Team Leader | 15 000 | 22 000 |
| Hardware Electronic Engineer    | 9 500  | 14 000 |

|                                      | 2022   |        |
|--------------------------------------|--------|--------|
|                                      | min    | max    |
| Embedded Software                    |        |        |
| Engineering Test Manager             | 19 000 | 27 000 |
| Engineering Test Team Leader         | 15 000 | 21 000 |
| Embedded Tester                      | 9 500  | 15 000 |
| Embedded Software Department Manager | 20 000 | 28 000 |
| Embedded Developer C/C++             | 10 000 | 18 000 |
| Embedded Software Team Leader        | 15 000 | 22 000 |

|                                         | 2022   |        |
|-----------------------------------------|--------|--------|
|                                         | min    | max    |
| Software                                |        |        |
| Software Development Department Manager | 21 000 | 29 000 |
| Software Development Team Leader        | 18 000 | 24 000 |
| Software Developer .Net                 | 12 000 | 17 000 |
| Software Developer Java                 | 12 000 | 18 000 |
| Software Developer Front-End            | 11 000 | 17 000 |
| Software Tester                         | 10 000 | 16 000 |





## Chemical and FMCG





## Chemical and FMCG

2022

|                                   | Lower Silesia |        | Kuyavia-Pomerania |        | Lublin |        | Lesser Poland |        | Masovia |        | Pomerania |        | Silesia |        | Greater Poland |        | West Pomerania |        | Łódź   |        |
|-----------------------------------|---------------|--------|-------------------|--------|--------|--------|---------------|--------|---------|--------|-----------|--------|---------|--------|----------------|--------|----------------|--------|--------|--------|
|                                   | min           | max    | min               | max    | min    | max    | min           | max    | min     | max    | min       | max    | min     | max    | min            | max    | min            | max    | min    | max    |
| Plant Manager                     | 20 000        | 30 000 | 18 000            | 28 000 | 17 000 | 27 000 | 19 000        | 33 000 | 24 000  | 35 000 | 20 000    | 30 000 | 18 000  | 32 000 | 19 000         | 30 000 | 18 000         | 28 000 | 20 000 | 30 000 |
| Department Manager                | 10 000        | 16 000 | 10 000            | 15 000 | 10 000 | 14 000 | 10 000        | 18 000 | 11 000  | 18 000 | 10 000    | 15 000 | 11 000  | 18 000 | 10 000         | 17 000 | 10 000         | 15 000 | 11 000 | 16 000 |
| Production Manager                | 13 000        | 18 000 | 11 000            | 16 000 | 12 000 | 17 000 | 13 000        | 19 000 | 15 000  | 20 000 | 12 000    | 18 000 | 12 000  | 19 000 | 13 000         | 19 000 | 11 000         | 17 000 | 12 000 | 18 000 |
| Manufacturing Engineer            | 6 500         | 9 500  | 6 500             | 9 000  | 6 500  | 9 000  | 7 000         | 9 000  | 7 000   | 10 000 | 6 500     | 9 500  | 7 000   | 9 000  | 6 500          | 9 000  | 6 500          | 9 000  | 7 000  | 9 000  |
| Technical Manager (Maintenance)   | 11 000        | 16 000 | 10 000            | 15 000 | 10 000 | 15 000 | 11 000        | 16 000 | 13 000  | 18 000 | 10 000    | 16 000 | 11 000  | 16 000 | 11 000         | 15 000 | 10 000         | 15 000 | 11 000 | 16 000 |
| Maintenance Engineer              | 6 500         | 9 500  | 6 500             | 8 500  | 6 000  | 9 000  | 7 000         | 9 000  | 7 500   | 9 000  | 6 500     | 9 000  | 7 500   | 10 000 | 7 000          | 9 500  | 6 500          | 8 500  | 7 000  | 9 000  |
| Mechanic                          | 5 500         | 7 000  | 5 000             | 6 500  | 5 000  | 6 500  | 5 000         | 6 500  | 6 000   | 7 500  | 4 500     | 7 000  | 5 000   | 6 500  | 5 000          | 6 000  | 5 000          | 6 500  | 5 000  | 6 500  |
| Mechanical Engineer               | 7 000         | 9 000  | 6 000             | 8 000  | 6 000  | 8 000  | 6 000         | 8 500  | 7 000   | 9 000  | 6 000     | 9 000  | 6 500   | 9 000  | 6 500          | 8 500  | 6 000          | 8 000  | 6 000  | 8 000  |
| Electrical Engineer               | 6 500         | 8 500  | 6 000             | 8 000  | 6 000  | 8 000  | 6 500         | 9 000  | 6 500   | 9 000  | 6 000     | 8 500  | 7 500   | 9 500  | 6 000          | 8 500  | 6 000          | 8 000  | 6 000  | 8 000  |
| Electrician                       | 5 500         | 7 000  | 5 500             | 7 000  | 5 500  | 7 000  | 5 000         | 7 000  | 5 000   | 7 000  | 5 000     | 7 000  | 5 500   | 7 500  | 5 500          | 7 000  | 5 500          | 7 000  | 5 500  | 7 000  |
| Automation Engineer               | 6 500         | 10 000 | 6 500             | 8 500  | 6 500  | 8 500  | 7 000         | 9 000  | 7 000   | 9 000  | 7 000     | 10 000 | 7 000   | 10 000 | 7 500          | 9 500  | 6 500          | 8 500  | 6 500  | 9 000  |
| Automation Technician             | 5 500         | 7 000  | 5 500             | 7 000  | 5 500  | 7 000  | 5 500         | 7 000  | 5 500   | 7 000  | 5 500     | 7 000  | 5 500   | 7 000  | 5 500          | 7 000  | 5 500          | 7 000  | 5 500  | 7 000  |
| Manufacturing Shift Manager       | 6 500         | 9 000  | 6 500             | 8 500  | 6 500  | 8 500  | 6 500         | 8 500  | 8 000   | 10 000 | 6 500     | 9 000  | 7 000   | 10 000 | 6 500          | 9 000  | 6 500          | 8 500  | 6 500  | 8 500  |
| Foreman                           | 4 500         | 6 500  | 4 500             | 6 500  | 4 500  | 6 500  | 4 500         | 6 500  | 5 000   | 6 500  | 5 000     | 6 500  | 5 500   | 6 500  | 5 500          | 6 500  | 4 500          | 6 500  | 5 000  | 6 500  |
| Qualified Production Worker       | 3 800         | 5 000  | 3 500             | 4 500  | 3 500  | 4 500  | 3 800         | 4 800  | 3 800   | 5 000  | 4 000     | 5 000  | 3 800   | 5 000  | 4 000          | 5 000  | 3 500          | 4 500  | 3 500  | 4 800  |
| Production Worker (non qualified) | 3 200         | 3 800  | 3 200             | 3 600  | 3 200  | 3 600  | 3 300         | 4 000  | 3 500   | 4 000  | 3 300     | 4 000  | 3 300   | 4 000  | 3 300          | 4 000  | 3 200          | 3 600  | 3 300  | 3 800  |
| Project Engineer                  | 6 500         | 10 000 | 6 000             | 8 500  | 6 000  | 8 500  | 6 500         | 9 000  | 8 000   | 11 000 | 7 000     | 9 000  | 7 500   | 9 500  | 7 000          | 9 000  | 6 000          | 8 500  | 6 500  | 9 000  |
| Project Manager                   | 11 000        | 15 000 | 10 000            | 14 000 | 10 000 | 14 000 | 11 000        | 15 000 | 11 000  | 18 000 | 10 000    | 15 000 | 12 000  | 18 000 | 12 000         | 18 000 | 10 000         | 14 000 | 10 000 | 14 000 |





## Chemical and FMCG

2022

|                                       | Lower Silesia |        | Kuyavia-Pomerania |        | Lublin |        | Lesser Poland |        | Masovia |        | Pomerania |        | Silesia |        | Greater Poland |        | West Pomerania |        | Łódź   |        |
|---------------------------------------|---------------|--------|-------------------|--------|--------|--------|---------------|--------|---------|--------|-----------|--------|---------|--------|----------------|--------|----------------|--------|--------|--------|
|                                       | min           | max    | min               | max    | min    | max    | min           | max    | min     | max    | min       | max    | min     | max    | min            | max    | min            | max    | min    | max    |
| Quality Manager                       | 13 000        | 18 000 | 12 000            | 16 000 | 12 000 | 16 000 | 13 000        | 21 000 | 15 000  | 21 000 | 12 000    | 18 000 | 13 000  | 20 000 | 12 000         | 18 000 | 12 000         | 16 000 | 11 000 | 17 000 |
| Quality Engineer                      | 6 000         | 9 500  | 5 500             | 8 000  | 5 500  | 8 000  | 7 000         | 9 000  | 7 000   | 9 000  | 6 000     | 8 500  | 7 000   | 9 000  | 6 500          | 9 000  | 5 500          | 8 500  | 6 500  | 8 500  |
| Process Engineer                      | 6 000         | 9 000  | 6 000             | 8 000  | 6 000  | 8 000  | 6 500         | 9 000  | 7 000   | 9 000  | 6 000     | 9 000  | 7 000   | 9 500  | 7 000          | 9 000  | 6 000          | 8 000  | 6 000  | 8 500  |
| Lean Manufacturing Manager            | 11 000        | 18 000 | 10 000            | 14 000 | 10 000 | 14 000 | 11 000        | 19 000 | 13 000  | 20 000 | 11 000    | 18 000 | 12 000  | 18 000 | 12 000         | 18 000 | 10 000         | 15 000 | 11 000 | 15 000 |
| Lean Manufacturing Specialist         | 6 500         | 9 500  | 6 000             | 8 000  | 6 000  | 8 000  | 6 500         | 9 500  | 7 000   | 10 000 | 6 500     | 9 500  | 7 000   | 9 500  | 7 000          | 10 000 | 6 000          | 8 000  | 6 500  | 9 000  |
| Design Engineer                       | 6 000         | 9 000  | 5 500             | 8 000  | 5 500  | 8 000  | 6 500         | 9 000  | 8 000   | 10 000 | 6 000     | 9 000  | 6 500   | 10 000 | 7 000          | 10 500 | 6 000          | 8 000  | 6 000  | 8 000  |
| Development Engineer                  | 7 000         | 10 000 | 6 500             | 8 500  | 6 500  | 8 500  | 7 500         | 10 000 | 8 000   | 11 000 | 7 000     | 9 500  | 8 000   | 11 000 | 7 000          | 10 000 | 6 500          | 8 500  | 7 000  | 9 000  |
| Quality Inspector                     | 3 500         | 4 500  | 3 500             | 4 500  | 3 500  | 4 500  | 3 500         | 4 800  | 4 000   | 5 000  | 3 500     | 4 500  | 3 500   | 4 800  | 3 500          | 4 800  | 3 500          | 4 500  | 3 500  | 4 500  |
| Laboratory Manager                    | 9 500         | 14 000 | 8 500             | 12 000 | 8 500  | 12 000 | 10 000        | 15 000 | 10 000  | 15 000 | 9 000     | 14 000 | 10 000  | 15 000 | 10 000         | 15 000 | 9 000          | 13 000 | 9 500  | 13 000 |
| Laboratory Technician                 | 4 500         | 6 500  | 4 000             | 5 500  | 4 000  | 5 500  | 5 000         | 6 000  | 5 000   | 6 000  | 4 500     | 6 000  | 5 000   | 6 000  | 5 000          | 6 000  | 4 000          | 5 500  | 4 500  | 5 500  |
| Chemical Engineer/laboratory Engineer | 6 000         | 8 500  | 5 500             | 7 500  | 5 500  | 8 000  | 6 500         | 8 500  | 6 500   | 9 000  | 6 000     | 8 000  | 6 500   | 8 500  | 6 500          | 8 500  | 6 000          | 7 500  | 5 500  | 7 500  |
| Chief Technologist                    | 9 000         | 13 000 | 8 000             | 12 000 | 8 500  | 11 500 | 9 500         | 15 000 | 10 000  | 15 000 | 9 000     | 13 000 | 10 000  | 15 000 | 10 000         | 14 000 | 8 500          | 12 000 | 8 500  | 12 000 |
| Logistics Manager                     | 12 000        | 16 000 | 11 000            | 15 000 | 11 000 | 15 000 | 12 000        | 17 000 | 12 000  | 19 000 | 12 000    | 16 000 | 12 000  | 18 000 | 12 000         | 17 000 | 11 000         | 15 000 | 12 000 | 15 000 |
| Logistics Specialist                  | 6 000         | 8 500  | 5 500             | 7 500  | 5 500  | 7 500  | 6 000         | 8 500  | 7 000   | 9 000  | 6 000     | 8 500  | 6 500   | 8 500  | 6 500          | 8 500  | 5 500          | 7 500  | 5 500  | 7 500  |
| Warehouse Manager                     | 9 000         | 13 000 | 8 500             | 12 000 | 8 500  | 12 000 | 9 000         | 14 000 | 10 000  | 14 000 | 8 500     | 13 000 | 9 500   | 13 000 | 9 000          | 13 000 | 8 500          | 12 000 | 9 000  | 14 000 |
| Warehouse Worker                      | 3 500         | 4 500  | 3 300             | 4 300  | 3 300  | 4 300  | 3 500         | 4 500  | 3 800   | 4 800  | 3 500     | 4 500  | 3 500   | 4 500  | 3 500          | 4 500  | 3 500          | 4 500  | 3 500  | 4 500  |
| Forklift Driver                       | 3 800         | 5 000  | 3 800             | 4 800  | 3 800  | 4 800  | 4 000         | 5 000  | 4 500   | 5 500  | 3 500     | 5 000  | 4 000   | 5 000  | 4 000          | 5 000  | 4 300          | 5 500  | 4 000  | 5 000  |
| Purchasing Manager                    | 12 000        | 19 000 | 12 000            | 16 000 | 12 000 | 16 000 | 13 000        | 19 000 | 16 000  | 22 000 | 10 000    | 16 000 | 14 000  | 21 000 | 13 000         | 19 000 | 11 000         | 16 000 | 13 000 | 18 000 |
| Strategic Buyer                       | 11 000        | 15 000 | 10 000            | 13 000 | 10 000 | 13 000 | 12 000        | 16 000 | 12 000  | 17 000 | 9 000     | 15 000 | 12 000  | 16 000 | 12 000         | 16 000 | 8 000          | 14 000 | 10 000 | 14 000 |
| Buyer                                 | 8 000         | 11 000 | 6 500             | 8 500  | 6 500  | 8 500  | 8 000         | 11 000 | 8 000   | 12 000 | 7 000     | 11 000 | 8 000   | 12 000 | 7 000          | 11 000 | 6 500          | 9 000  | 7 000  | 9 500  |
| Production Planning Specialist        | 6 000         | 8 500  | 5 500             | 7 500  | 5 500  | 7 500  | 6 000         | 8 500  | 7 000   | 9 500  | 5 500     | 7 500  | 6 500   | 9 000  | 6 000          | 8 500  | 5 500          | 7 000  | 6 000  | 8 000  |





# Automotive and Appliances







## Automotive and Appliances

2022

|                                   | Lower Silesia |        | Opole  |        | Lublin |        | Łódź   |        | Lesser Poland |        | Masovia |        | Subcarpathia |        | Silesia |        | Świętokrzyskie |        | Greater Poland |        | Pomerania |        |
|-----------------------------------|---------------|--------|--------|--------|--------|--------|--------|--------|---------------|--------|---------|--------|--------------|--------|---------|--------|----------------|--------|----------------|--------|-----------|--------|
|                                   | min           | max    | min    | max    | min    | max    | min    | max    | min           | max    | min     | max    | min          | max    | min     | max    | min            | max    | min            | max    | min       | max    |
| Plant Manager                     | 20 000        | 35 000 | 20 000 | 35 000 | 19 000 | 33 000 | 20 000 | 33 000 | 18 000        | 35 000 | 28 000  | 37 000 | 17 000       | 30 000 | 18 000  | 35 000 | 17 000         | 30 000 | 25 000         | 37 000 | 20 000    | 35 000 |
| Department Manager                | 10 000        | 19 000 | 10 000 | 19 000 | 10 000 | 15 000 | 10 000 | 15 000 | 10 500        | 18 000 | 11 000  | 18 000 | 9 500        | 14 000 | 11 000  | 18 000 | 9 500          | 14 000 | 11 000         | 18 000 | 10 000    | 19 000 |
| Production Manager                | 13 000        | 18 000 | 13 000 | 18 000 | 12 000 | 17 000 | 11 000 | 16 000 | 10 000        | 18 000 | 16 000  | 20 000 | 9 000        | 15 000 | 13 000  | 19 000 | 11 000         | 18 000 | 14 000         | 20 000 | 13 000    | 18 000 |
| Manufacturing Engineer            | 6 000         | 10 000 | 6 000  | 10 000 | 5 500  | 8 500  | 6 000  | 8 500  | 6 500         | 9 000  | 7 000   | 9 500  | 6 000        | 8 000  | 7 000   | 9 000  | 5 500          | 8 000  | 7 000          | 9 500  | 6 000     | 10 000 |
| Technical Manager (Maintenance)   | 11 000        | 16 000 | 11 000 | 16 000 | 10 000 | 14 000 | 10 000 | 15 000 | 10 000        | 16 000 | 13 000  | 17 000 | 10 000       | 14 000 | 11 000  | 16 000 | 10 000         | 14 000 | 11 000         | 15 000 | 11 000    | 16 000 |
| Maintenance Engineer              | 6 500         | 9 500  | 6 500  | 9 500  | 6 000  | 9 000  | 6 500  | 8 500  | 7 000         | 9 000  | 7 500   | 9 000  | 6 000        | 8 000  | 7 000   | 10 000 | 6 000          | 9 000  | 6 700          | 9 500  | 6 000     | 9 000  |
| Maintenance Technician            | 5 500         | 7 500  | 5 000  | 7 500  | 5 600  | 7 000  | 4 500  | 6 500  | 5 000         | 6 500  | 5 000   | 7 000  | 4 800        | 6 500  | 5 000   | 7 000  | 4 500          | 6 000  | 5 000          | 7 000  | 5 000     | 7 500  |
| Mechanical Engineer               | 7 000         | 9 000  | 6 000  | 9 000  | 6 000  | 8 000  | 6 000  | 8 000  | 6 000         | 8 500  | 7 000   | 9 000  | 5 500        | 7 500  | 6 500   | 9 000  | 6 000          | 8 000  | 6 200          | 8 500  | 6 000     | 9 000  |
| Mechanic                          | 5 500         | 7 000  | 4 500  | 7 000  | 5 000  | 7 000  | 5 000  | 6 500  | 5 000         | 6 500  | 6 000   | 7 500  | 4 800        | 6 200  | 5 000   | 6 500  | 4 500          | 6 000  | 5 000          | 6 000  | 4 500     | 7 000  |
| Electrical Engineer               | 6 500         | 8 500  | 6 000  | 8 500  | 6 000  | 8 000  | 6 000  | 8 000  | 6 500         | 9 000  | 6 500   | 9 000  | 6 000        | 8 000  | 7 500   | 9 500  | 6 000          | 8 000  | 6 000          | 8 500  | 6 000     | 8 500  |
| Electrician                       | 5 500         | 7 000  | 5 000  | 7 000  | 5 500  | 7 000  | 5 500  | 7 000  | 5 000         | 7 000  | 5 000   | 7 000  | 4 800        | 6 000  | 5 500   | 7 500  | 4 800          | 6 000  | 5 500          | 7 000  | 5 000     | 7 000  |
| Automation Engineer               | 6 500         | 10 000 | 6 500  | 10 000 | 6 300  | 8 500  | 6 500  | 8 500  | 7 000         | 9 000  | 7 000   | 9 000  | 6 300        | 8 800  | 7 000   | 10 000 | 6 500          | 8 500  | 7 500          | 9 500  | 7 000     | 10 000 |
| Automation Technician             | 5 500         | 7 000  | 5 500  | 7 000  | 5 500  | 7 000  | 5 500  | 7 000  | 5 500         | 7 000  | 5 500   | 7 000  | 5 000        | 6 000  | 5 500   | 7 000  | 4 800          | 6 000  | 5 500          | 7 000  | 5 500     | 7 000  |
| Workshop Manager                  | 8 000         | 12 000 | 8 000  | 12 000 | 8 000  | 11 000 | 8 000  | 11 000 | 8 500         | 13 000 | 9 000   | 12 000 | 8 500        | 11 000 | 9 500   | 13 000 | 9 000          | 11 000 | 9 000          | 13 000 | 8 000     | 12 000 |
| Tooling Engineer                  | 6 500         | 8 500  | 6 500  | 8 500  | 6 000  | 8 000  | 6 000  | 8 000  | 6 500         | 8 500  | 7 000   | 9 000  | 6 000        | 8 000  | 7 000   | 9 000  | 6 000          | 8 000  | 6 500          | 9 000  | 6 000     | 8 500  |
| Manufacturing Shift Manager       | 6 500         | 9 000  | 6 500  | 9 000  | 6 500  | 8 500  | 6 500  | 8 500  | 6 500         | 8 500  | 8 000   | 10 000 | 6 500        | 8 000  | 7 000   | 10 000 | 6 000          | 8 000  | 6 500          | 9 000  | 6 500     | 9 000  |
| Foreman                           | 4 500         | 6 500  | 4 500  | 6 500  | 4 500  | 6 500  | 4 500  | 6 500  | 4 500         | 6 500  | 5 000   | 6 500  | 4 500        | 6 000  | 5 500   | 6 500  | 4 500          | 6 000  | 5 000          | 6 500  | 5 000     | 6 500  |
| Assembly Line Worker              | 3 500         | 4 500  | 3 500  | 4 500  | 3 300  | 4 300  | 3 300  | 4 500  | 3 500         | 4 500  | 3 700   | 4 600  | 3 500        | 4 200  | 3 500   | 4 500  | 3 300          | 4 200  | 3 500          | 4 500  | 3 500     | 4 500  |
| Production Operator               | 3 800         | 5 000  | 3 800  | 5 000  | 3 500  | 4 500  | 3 500  | 4 800  | 3 800         | 4 800  | 3 800   | 5 000  | 3 700        | 4 800  | 3 800   | 5 000  | 3 500          | 4 800  | 4 000          | 5 000  | 4 000     | 5 000  |
| Production Worker (non qualified) | 3 200         | 3 800  | 3 200  | 3 800  | 3 200  | 3 600  | 3 300  | 3 800  | 3 300         | 4 000  | 3 500   | 4 000  | 3 300        | 3 800  | 3 300   | 4 000  | 3 300          | 3 800  | 3 300          | 4 000  | 3 300     | 4 000  |





## Automotive and Appliances

|                             | 2022          |        |        |        |        |        |        |        |               |        |         |        |              |        |         |        |                |        |                |        |           |        |
|-----------------------------|---------------|--------|--------|--------|--------|--------|--------|--------|---------------|--------|---------|--------|--------------|--------|---------|--------|----------------|--------|----------------|--------|-----------|--------|
|                             | Lower Silesia |        | Opole  |        | Lublin |        | Łódź   |        | Lesser Poland |        | Masovia |        | Subcarpathia |        | Silesia |        | Świętokrzyskie |        | Greater Poland |        | Pomerania |        |
|                             | min           | max    | min    | max    | min    | max    | min    | max    | min           | max    | min     | max    | min          | max    | min     | max    | min            | max    | min            | max    | min       | max    |
| CNC Programmer              | 5 000         | 8 000  | 5 000  | 7 000  | 4 800  | 6 500  | 5 000  | 7 000  | 5 000         | 7 000  | 5 500   | 7 500  | 5 000        | 6 500  | 5 000   | 7 000  | 5 000          | 6 500  | 5 000          | 7 000  | 5 000     | 8 000  |
| CNC Operator                | 4 000         | 5 500  | 4 000  | 5 500  | 3 800  | 4 800  | 3 800  | 5 000  | 4 000         | 5 000  | 4 300   | 5 500  | 3 800        | 5 500  | 4 000   | 5 500  | 3 800          | 5 500  | 4 000          | 5 200  | 4 000     | 5 000  |
| Master Welder Engineer      | 10 000        | 14 000 | 10 000 | 14 000 | 10 000 | 12 000 | 9 000  | 14 000 | 10 000        | 14 000 | 11 000  | 14 000 | 9 000        | 14 000 | 10 000  | 15 000 | 9 000          | 14 000 | 10 000         | 14 000 | 10 000    | 14 000 |
| Welder (tig, mig, mag etc)  | 5 000         | 8 000  | 5 000  | 7 500  | 5 500  | 7 500  | 5 000  | 6 500  | 5 500         | 7 500  | 5 500   | 7 500  | 5 500        | 7 500  | 5 500   | 7 500  | 5 000          | 7 000  | 5 000          | 7 500  | 5 000     | 8 000  |
| Project Engineer            | 6 500         | 10 000 | 6 500  | 10 000 | 6 000  | 8 500  | 6 500  | 9 000  | 6 500         | 9 000  | 8 000   | 11 000 | 6 500        | 8 500  | 7 500   | 9 500  | 6 500          | 8 500  | 7 000          | 9 000  | 7 000     | 9 000  |
| Project Manager             | 11 000        | 15 000 | 11 000 | 15 000 | 10 000 | 14 000 | 10 000 | 14 000 | 11 000        | 15 000 | 11 000  | 18 000 | 10 000       | 14 000 | 12 000  | 18 000 | 10 000         | 14 000 | 10 000         | 15 000 | 10 000    | 15 000 |
| Program Manager             | 15 000        | 19 000 | 13 000 | 17 000 | 12 000 | 16 000 | 12 000 | 16 000 | 13 000        | 19 000 | 14 000  | 20 000 | 12 000       | 16 000 | 14 000  | 20 000 | 11 000         | 15 500 | 12 000         | 18 000 | 12 000    | 19 000 |
| Quality Manager             | 12 000        | 19 000 | 12 000 | 19 000 | 11 000 | 17 000 | 11 000 | 17 000 | 12 000        | 18 000 | 15000   | 20 000 | 11 000       | 17 000 | 12 000  | 18 000 | 11 000         | 17 000 | 12 000         | 19 000 | 12 000    | 19 000 |
| Quality Engineer            | 6 000         | 10 000 | 5 500  | 10 000 | 5 500  | 8 500  | 6 000  | 8 500  | 6 500         | 8 500  | 6 500   | 9 000  | 6 000        | 8 500  | 6 500   | 9 000  | 6 000          | 8 000  | 7 000          | 9 000  | 6 000     | 9 500  |
| Customer QA Engineer        | 7 000         | 10 000 | 7 000  | 10 000 | 6 000  | 8 500  | 7 000  | 8 500  | 7 000         | 9 000  | 8 000   | 9 000  | 6 500        | 8 500  | 7 000   | 9 500  | 6 500          | 8 500  | 7 000          | 9 000  | 6 500     | 9 500  |
| Supplier QA Manager         | 11 000        | 15 000 | 11 000 | 15 000 | 10 000 | 14 000 | 10 000 | 14 000 | 10 000        | 15 000 | 12 000  | 16 000 | 10 000       | 14 000 | 12 000  | 16 000 | 10 000         | 14 000 | 10 000         | 17 000 | 10 000    | 16 000 |
| Supplier QA Engineer        | 7 000         | 10 000 | 7 000  | 10 000 | 6 500  | 8 500  | 7 000  | 9 000  | 7 500         | 9 500  | 7 500   | 9 500  | 6 500        | 8 500  | 7 000   | 9 500  | 6 500          | 9 000  | 7 000          | 10 000 | 7 000     | 10 000 |
| Quality Technician          | 4 500         | 6 500  | 4 500  | 6 500  | 4 000  | 5 000  | 4 200  | 5 500  | 4 500         | 5 500  | 4 500   | 5 500  | 4 000        | 5 000  | 4 500   | 5 500  | 4 000          | 5 000  | 4 500          | 5 500  | 4 500     | 5 500  |
| Quality Inspector           | 3 500         | 4 500  | 3 500  | 4 500  | 3 500  | 4 500  | 3 500  | 4 500  | 3 500         | 4 800  | 4 000   | 5 000  | 3 500        | 4 500  | 3 500   | 4 800  | 3 500          | 4 500  | 3 500          | 4 800  | 3 500     | 4 500  |
| Process Engineering Manager | 11 000        | 16 000 | 11 000 | 16 000 | 10 000 | 14 000 | 11 000 | 15 000 | 11 000        | 16 000 | 11 000  | 16 000 | 10 000       | 14 000 | 12 000  | 16 000 | 10 000         | 14 000 | 11 000         | 16 000 | 11 000    | 16 000 |
| Design Engineer             | 6 000         | 9 000  | 6 000  | 9 000  | 5 500  | 8 000  | 6 000  | 8 000  | 6 500         | 9 000  | 8 000   | 10 000 | 6 000        | 8 000  | 6 500   | 10 000 | 5 500          | 8 000  | 7 000          | 11 000 | 6 000     | 9 000  |
| Process Engineer            | 6 000         | 9 500  | 6 000  | 9 500  | 6 000  | 8 000  | 6 000  | 8 500  | 6 000         | 8 500  | 7 000   | 9 000  | 6 000        | 8 000  | 7 000   | 9 500  | 6 000          | 8 000  | 7 000          | 9 000  | 6 000     | 9 000  |
| Technologist                | 5 500         | 8 500  | 5 500  | 8 500  | 5 500  | 8 000  | 6 000  | 8 500  | 5 500         | 8 500  | 7 000   | 9 000  | 5 500        | 8 000  | 6 000   | 9 000  | 5 500          | 8 000  | 6 500          | 9 000  | 5 500     | 8 500  |
| Laboratory Manager          | 10 000        | 13 000 | 10 000 | 13 000 | 8 500  | 11 000 | 9 000  | 12 000 | 9 500         | 13 000 | 10 000  | 14 000 | 8 500        | 10 500 | 10 000  | 14 000 | 8 500          | 10 500 | 10 000         | 13 000 | 10 000    | 12 000 |
| Industrialization Engineer  | 7 500         | 11 000 | 7 500  | 11 000 | 7 000  | 9 000  | 7 000  | 9 000  | 8 000         | 10 000 | 8 000   | 10 000 | 7 500        | 9 000  | 8 000   | 10 000 | 7 500          | 9 000  | 7 500          | 10 000 | 7 500     | 10 000 |



## Automotive and Appliances

2022

|                               | Lower Silesia |        | Opole  |        | Lublin |        | Łódź   |        | Lesser Poland |        | Masovia |        | Subcarpathia |        | Silesia |        | Świętokrzyskie |        | Greater Poland |        | Pomerania |        |
|-------------------------------|---------------|--------|--------|--------|--------|--------|--------|--------|---------------|--------|---------|--------|--------------|--------|---------|--------|----------------|--------|----------------|--------|-----------|--------|
|                               | min           | max    | min    | max    | min    | max    | min    | max    | min           | max    | min     | max    | min          | max    | min     | max    | min            | max    | min            | max    | min       | max    |
| Lean Manufacturing Manager    | 11 000        | 18 000 | 11 000 | 18 000 | 10 000 | 14 000 | 11 000 | 15 000 | 10 000        | 18 000 | 13 000  | 20 000 | 10 000       | 14 000 | 12 000  | 18 000 | 10 000         | 14 000 | 12 000         | 18 000 | 11 000    | 18 000 |
| Lean Manufacturing Specialist | 6 500         | 9 500  | 6 500  | 9 500  | 6 000  | 8 000  | 6 500  | 9 000  | 6 500         | 9 500  | 7 000   | 10 000 | 6 000        | 8 000  | 7 000   | 9 500  | 6 500          | 8 000  | 7 000          | 10 000 | 6 500     | 9 500  |
| Logistics Director            | 16 000        | 24 000 | 16 000 | 24 000 | 15 000 | 20 000 | 16 000 | 22 000 | 16 000        | 24 000 | 20 000  | 28 000 | 15 000       | 20 000 | 17 000  | 25 000 | 15 000         | 20 000 | 17 000         | 25 000 | 16 000    | 24 000 |
| Logistics Manager             | 12 000        | 16 000 | 12 000 | 16 000 | 11 000 | 15 000 | 12 000 | 15 000 | 12 000        | 17 000 | 12 000  | 19 000 | 10 000       | 15 000 | 12 000  | 18 000 | 10 000         | 15 000 | 12 000         | 17 000 | 12 000    | 16 000 |
| Logistics Specialist          | 6 000         | 8 500  | 6 000  | 8 500  | 5 500  | 7 500  | 5 500  | 7 500  | 6 000         | 8 000  | 7 000   | 9 000  | 5 500        | 7 500  | 6 500   | 8 500  | 5 500          | 7 500  | 6 500          | 8 500  | 6 000     | 8 500  |
| Warehouse Manager             | 9 000         | 13 000 | 9 000  | 13 000 | 8 500  | 12 000 | 9 000  | 13 000 | 9 000         | 13 000 | 10 000  | 14 000 | 8 500        | 12 000 | 9 500   | 13 000 | 8 500          | 12 000 | 9 000          | 13 000 | 9 000     | 13 000 |
| Warehouse Worker              | 3 500         | 4 500  | 3 500  | 4 500  | 3 300  | 4 300  | 3 500  | 4 500  | 3 500         | 4 500  | 3 800   | 4 800  | 3 300        | 4 300  | 3 500   | 4 500  | 3 300          | 4 300  | 3 500          | 4 500  | 3 300     | 4 300  |
| Forklift Driver               | 3 800         | 5 000  | 3 800  | 5 000  | 3 800  | 4 800  | 4 000  | 5 000  | 4 000         | 5 000  | 4 500   | 5 500  | 3 800        | 4 800  | 4 000   | 5 000  | 3 800          | 4 800  | 4 000          | 5 000  | 4 000     | 5 000  |
| Purchasing Manager            | 12 000        | 19 000 | 12 000 | 19 000 | 12 000 | 16 000 | 13 000 | 18 000 | 13 000        | 19 000 | 16 000  | 22 000 | 11 000       | 16 000 | 14 000  | 21 000 | 11 000         | 16 000 | 13 000         | 19 000 | 13 000    | 18 000 |
| Strategic Buyer               | 11 000        | 15 000 | 11 000 | 15 000 | 10 000 | 13 000 | 10 000 | 14 000 | 12 000        | 16 000 | 12 000  | 17 000 | 10 000       | 14 500 | 12 000  | 16 000 | 10 000         | 14 000 | 12 000         | 16 000 | 10 000    | 14 000 |
| Buyer                         | 7 000         | 11 000 | 7 000  | 11 000 | 6 500  | 8 000  | 7 000  | 9 500  | 7 500         | 11 000 | 8 000   | 12 000 | 6 500        | 8 500  | 7 500   | 12 000 | 6 500          | 8 500  | 7 000          | 11 000 | 7 000     | 11 000 |
| Production Planing Specialist | 6 000         | 8 500  | 6 000  | 8 500  | 5 500  | 7 500  | 6 000  | 8 000  | 6 000         | 8 500  | 7 000   | 9 500  | 5 500        | 7 500  | 6 500   | 9 000  | 5 500          | 7 500  | 6 000          | 8 500  | 6 000     | 8 000  |





## Back Office





Back Office

2022

|  | Lower Silesia |     | Kuyavia-Pomerania |     | Lublin |     | Łódź |     | Lubusz |     | Lesser Poland |     | Masovia |     | Opole |     | Subcarpathia |     | Pomerania |     | Silesia |     | Świętokrzyskie |     | Greater Poland |     | West Pomerania |     |
|--|---------------|-----|-------------------|-----|--------|-----|------|-----|--------|-----|---------------|-----|---------|-----|-------|-----|--------------|-----|-----------|-----|---------|-----|----------------|-----|----------------|-----|----------------|-----|
|  | min           | max | min               | max | min    | max | min  | max | min    | max | min           | max | min     | max | min   | max | min          | max | min       | max | min     | max | min            | max | min            | max | min            | max |

F&A

|                            |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|----------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Finance Director           | 15 000 | 28 000 | 14 000 | 24 000 | 14 000 | 24 000 | 16 000 | 26 000 | 14 000 | 24 000 | 18 000 | 30 000 | 20 000 | 30 000 | 15 000 | 28 000 | 15 000 | 24 000 | 15 000 | 28 000 | 18 000 | 30 000 | 14 000 | 23 000 | 18 000 | 29 000 | 15 000 | 25 000 |
| Plant Financial Controller | 12 000 | 20 000 | 12 000 | 17 000 | 11 000 | 16 000 | 12 000 | 18 000 | 11 000 | 16 000 | 16 000 | 21 000 | 15 000 | 23 000 | 12 000 | 19 000 | 11 000 | 15 000 | 12 000 | 20 000 | 13 000 | 20 000 | 10 000 | 15 000 | 12 000 | 20 000 | 11 000 | 16 000 |
| Chief Accountant           | 12 000 | 17 000 | 11 000 | 15 000 | 10 000 | 14 000 | 10 000 | 16 000 | 10 000 | 15 000 | 13 000 | 18 000 | 12 000 | 18 000 | 10 000 | 17 000 | 10 000 | 14 000 | 11 000 | 17 000 | 12 000 | 18 000 | 10 000 | 13 000 | 12 000 | 18 000 | 11 000 | 15 000 |
| Accountant                 | 6 000  | 9 000  | 5 500  | 8 000  | 5 500  | 7 500  | 6 000  | 8 500  | 5 500  | 8 000  | 6 300  | 9 000  | 7 000  | 9 000  | 5 500  | 9 000  | 5 500  | 7 500  | 6 000  | 9 000  | 6 000  | 9 000  | 5 500  | 7 000  | 6 000  | 9 000  | 6 000  | 8 000  |
| Tax Specialist             | 6 000  | 10 000 | 6 000  | 9 000  | 6 000  | 8 000  | 7 000  | 9 000  | 7 000  | 9 000  | 7 500  | 10 000 | 8 000  | 12 000 | 6 000  | 10 000 | 6 000  | 8 000  | 6 000  | 10 000 | 7 000  | 11 000 | 6 000  | 8 500  | 6 500  | 10 000 | 6 500  | 9 000  |
| Auditor                    | 9 000  | 13 000 | 8 000  | 10 000 | 7 000  | 9 300  | 7 000  | 10 000 | 7 500  | 10 000 | 7 800  | 12 500 | 10 000 | 14 000 | 7 000  | 12 000 | 6 000  | 9 000  | 9 000  | 14 000 | 8 000  | 12 000 | 6 000  | 9 000  | 8 500  | 12 000 | 8 000  | 11 000 |
| Financial Analyst          | 7 000  | 12 000 | 7 000  | 11 000 | 7 000  | 9 000  | 7 000  | 10 000 | 7 000  | 11 000 | 8 000  | 12 000 | 8 000  | 12 000 | 7 000  | 11 000 | 6 300  | 9 500  | 7 000  | 12 000 | 8 500  | 12 500 | 6 300  | 9 000  | 8 000  | 12 000 | 7 000  | 10 000 |
| Controlling Specialist     | 7 000  | 10 000 | 7 000  | 10 000 | 6 000  | 8 000  | 7 000  | 10 000 | 7 000  | 10 000 | 8 000  | 10 000 | 8 500  | 10 500 | 7 000  | 10 000 | 6 000  | 8 400  | 7 000  | 10 000 | 7 000  | 10 000 | 6 000  | 8 000  | 8 000  | 11 000 | 7 000  | 9 000  |
| Invoicing Clerk            | 4 500  | 5 500  | 4 000  | 5 000  | 3 800  | 5 000  | 4 500  | 5 500  | 4 000  | 5 000  | 4 500  | 6 000  | 5 000  | 6 000  | 4 000  | 5 500  | 4 000  | 5 000  | 4 500  | 5 500  | 4 500  | 5 500  | 4 000  | 5 000  | 4 500  | 5 500  | 4 000  | 4 500  |

HR

|                              |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| HR Director                  | 16 000 | 26 000 | 15 000 | 22 000 | 14 000 | 21 000 | 15 000 | 22 000 | 15 000 | 22 000 | 18 000 | 26 000 | 20 000 | 26 000 | 16 000 | 23 000 | 14 000 | 20 000 | 17 000 | 26 000 | 18 000 | 25 000 | 15 000 | 20 000 | 18 000 | 27 000 | 16 000 | 24 000 |
| HR Manager                   | 13 000 | 21 000 | 12 000 | 17 000 | 12 000 | 16 000 | 12 000 | 17 000 | 12 000 | 16 000 | 16 000 | 22 000 | 16 000 | 22 000 | 13 000 | 18 000 | 11 000 | 16 000 | 14 000 | 23 000 | 13 000 | 20 000 | 11 000 | 16 000 | 15 000 | 23 000 | 13 000 | 18 000 |
| HR BP                        | 8 000  | 13 000 | 8 000  | 12 000 | 8 000  | 12 000 | 8 000  | 12 000 | 8 000  | 12 000 | 9 000  | 14 000 | 10 000 | 14 000 | 8 000  | 13 000 | 8 000  | 11 000 | 8 500  | 13 000 | 8 000  | 13 000 | 8 000  | 11 000 | 8 000  | 13 000 | 8 000  | 12 000 |
| HR Specialist                | 5 500  | 8 000  | 5 000  | 8 000  | 5 000  | 7 000  | 5 500  | 7 500  | 5 000  | 7 500  | 6 300  | 8 500  | 6 500  | 8 500  | 5 000  | 8 000  | 5 500  | 7 500  | 5 500  | 8 000  | 6 000  | 8 000  | 5 500  | 7 000  | 6 000  | 8 000  | 5 500  | 8 000  |
| HR Administrator             | 4 500  | 6 000  | 4 500  | 5 500  | 4 200  | 5 500  | 4 200  | 5 500  | 4 200  | 5 500  | 5 000  | 6 500  | 5 000  | 7 000  | 4 200  | 6 000  | 4 500  | 5 500  | 4 500  | 5 500  | 4 500  | 6 000  | 4 500  | 5 500  | 4 500  | 6 000  | 4 500  | 6 000  |
| HR & Admnistration Manager   | 10 000 | 15 000 | 10 000 | 14 000 | 9 000  | 13 000 | 10 000 | 14 000 | 10 000 | 14 000 | 12 000 | 16 000 | 12 000 | 17 000 | 10 000 | 15 000 | 9 000  | 13 000 | 11 000 | 15 000 | 11 000 | 15 000 | 9 000  | 13 000 | 12 000 | 17 000 | 12 000 | 15 000 |
| Office Assistant             | 4 500  | 6 000  | 4 500  | 5 500  | 4 200  | 5 000  | 4 500  | 5 600  | 4 200  | 5 500  | 4 500  | 6 500  | 5 000  | 7 000  | 4 200  | 6 000  | 3 800  | 4 800  | 5 000  | 6 000  | 4 500  | 6 000  | 4 500  | 5 500  | 5 000  | 6 500  | 4 500  | 5 500  |
| Payroll Manager              | 8 500  | 14 000 | 8 000  | 12 000 | 8 000  | 12 000 | 8 000  | 12 000 | 8 000  | 12 000 | 10 000 | 14 000 | 11 000 | 15 000 | 8 000  | 12 000 | 8 000  | 12 000 | 10 000 | 13 000 | 9 000  | 13 000 | 8 000  | 12 000 | 10 000 | 15 000 | 10 000 | 14 000 |
| Payroll Specialist           | 5 500  | 9 000  | 5 500  | 7 500  | 5 000  | 6 500  | 5 000  | 6 000  | 5 000  | 6 500  | 6 000  | 8 500  | 7 000  | 9 000  | 5 500  | 8 000  | 5 000  | 6 500  | 5 300  | 8 000  | 5 500  | 8 000  | 5 500  | 7 000  | 5 500  | 8 000  | 5 500  | 7 500  |
| H&S Manager                  | 9 000  | 14 000 | 9 000  | 13 000 | 8 000  | 13 000 | 10 000 | 14 000 | 9 000  | 13 500 | 11 000 | 15 000 | 12 000 | 15 000 | 9 000  | 13 000 | 9 000  | 13 000 | 10 000 | 15 000 | 10 000 | 15 000 | 9 000  | 13 000 | 10 000 | 15 000 | 10 000 | 14 000 |
| Health and Safety Specialist | 6 000  | 8 000  | 5 000  | 8 000  | 5 000  | 7 000  | 6 000  | 8 000  | 6 000  | 7 000  | 6 300  | 8 500  | 6 500  | 8 500  | 5 000  | 8 000  | 5 500  | 7 000  | 5 300  | 8 500  | 6 000  | 8 500  | 5 500  | 7 500  | 6 000  | 8 500  | 5 000  | 8 000  |





Back Office

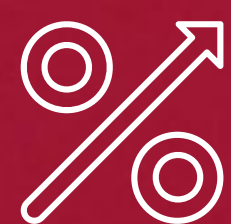
2022

|                       | Lower Silesia |        | Kuyavia-Pomerania |        | Lublin |        | Łódź   |        | Lubusz |        | Lesser Poland |        | Masovia |        | Opole  |        | Subcarpathia |        | Pomerania |        | Silesia |        | Świętokrzyskie |        | Greater Poland |        | West Pomerania |        |
|-----------------------|---------------|--------|-------------------|--------|--------|--------|--------|--------|--------|--------|---------------|--------|---------|--------|--------|--------|--------------|--------|-----------|--------|---------|--------|----------------|--------|----------------|--------|----------------|--------|
|                       | min           | max    | min               | max    | min    | max    | min    | max    | min    | max    | min           | max    | min     | max    | min    | max    | min          | max    | min       | max    | min     | max    | min            | max    | min            | max    | min            | max    |
| S&M                   |               |        |                   |        |        |        |        |        |        |        |               |        |         |        |        |        |              |        |           |        |         |        |                |        |                |        |                |        |
| Marketing Director    | 15 000        | 23 000 | 14 000            | 21 000 | 14 000 | 20 000 | 14 000 | 21 000 | 15 000 | 22 000 | 16 000        | 22 000 | 18 000  | 26 000 | 14 000 | 22 000 | 13 000       | 18 000 | 14 000    | 25 000 | 16 000  | 25 000 | 12 000         | 18 000 | 15 000         | 25 000 | 14 000         | 22 000 |
| Marketing Manager     | 12 000        | 18 000 | 11 000            | 17 000 | 11 000 | 16 000 | 11 000 | 16 000 | 11 000 | 16 000 | 13 000        | 18 000 | 15 000  | 18 000 | 12 000 | 17 000 | 11 000       | 14 000 | 12 000    | 20 000 | 12 000  | 18 000 | 11 000         | 14 000 | 14 000         | 19 000 | 12 000         | 16 000 |
| PR Manager            | 10 000        | 14 000 | 9 000             | 13 000 | 9 000  | 12 000 | 9 000  | 12 000 | 9 000  | 13 000 | 10 500        | 14 500 | 10 000  | 14 000 | 10 000 | 13 000 | 9 000        | 11 000 | 10 000    | 14 000 | 10 000  | 13 000 | 9 000          | 12 000 | 10 000         | 15 000 | 10 000         | 13 000 |
| Marketing Specialist  | 5 500         | 8 500  | 5 000             | 7 500  | 5 000  | 6 000  | 5 000  | 7 000  | 5 000  | 7 000  | 6 000         | 8 000  | 6 000   | 9 000  | 5 000  | 8 000  | 5 000        | 6 500  | 5 000     | 8 000  | 6 500   | 8 000  | 5 500          | 7 000  | 5 500          | 8 000  | 5 500          | 8 000  |
| Sales Director        | 15 000        | 25 000 | 15 000            | 22 000 | 15 000 | 22 000 | 15 000 | 23 000 | 15 000 | 22 000 | 18 000        | 25 000 | 18 000  | 26 000 | 15 000 | 22 000 | 15 000       | 21 000 | 15 000    | 22 000 | 18 000  | 25 000 | 15 000         | 21 000 | 18 000         | 26 000 | 15 000         | 23 000 |
| Sales Manager         | 11 000        | 21 000 | 11 000            | 16 000 | 10 000 | 15 000 | 10 000 | 16 000 | 12 000 | 16 000 | 12 000        | 18 000 | 14 000  | 19 000 | 10 000 | 18 000 | 10 000       | 15 000 | 11 000    | 16 000 | 14 000  | 19 000 | 11 000         | 15 000 | 14 000         | 18 000 | 13 000         | 17 000 |
| Sales Engineer        | 6 000         | 13 000 | 6 000             | 9 500  | 6 000  | 9 000  | 6 000  | 9 000  | 6 000  | 9 000  | 6 500         | 10 000 | 8 000   | 11 000 | 6 000  | 12 000 | 6 500        | 8 500  | 6 000     | 12 000 | 7 000   | 11 000 | 6 500          | 9 000  | 7 000          | 10 000 | 6 000          | 9 000  |
| RFQ Engineer          | 5 500         | 8 500  | 5 500             | 8 000  | 5 000  | 7 500  | 6 000  | 9 000  | 5 500  | 7 500  | 6 500         | 9 000  | 7 500   | 9 000  | 5 500  | 8 000  | 5 000        | 7 000  | 5 500     | 8 500  | 6 500   | 9 000  | 6 000          | 8 000  | 6 000          | 8 500  | 6 000          | 8 000  |
| Key Account Manager   | 8 000         | 12 000 | 7 000             | 11 000 | 6 500  | 9 500  | 7 000  | 11 000 | 7 000  | 11 000 | 8 000         | 12 000 | 11 000  | 14 000 | 7 000  | 11 000 | 7 000        | 9 500  | 8 000     | 11 000 | 8 000   | 12 000 | 6 500          | 9 500  | 8 000          | 12 000 | 8 000          | 10 000 |
| Sales Representative  | 5 000         | 7 500  | 5 000             | 6 000  | 5 000  | 6 000  | 5 000  | 7 000  | 5 000  | 7 000  | 5 500         | 8 000  | 6 000   | 8 500  | 5 000  | 7 000  | 5 000        | 7 000  | 5 000     | 8 000  | 6 000   | 8 000  | 5 500          | 7 000  | 6 000          | 8 000  | 5 500          | 7 000  |
| Technical Advisor     | 5 000         | 7 500  | 5 000             | 6 000  | 5 000  | 6 000  | 5 000  | 7 500  | 5 000  | 6 500  | 6 000         | 7 500  | 7 000   | 8 500  | 5 000  | 7 500  | 5 200        | 6 500  | 5 000     | 8 000  | 6 000   | 7 500  | 5 500          | 6 500  | 6 000          | 8 000  | 5 000          | 7 000  |
| Service Manager       | 10 000        | 13 000 | 9 000             | 13 000 | 9 000  | 13 000 | 9 000  | 12 000 | 8 500  | 13 500 | 10 000        | 15 000 | 13 000  | 16 000 | 9 000  | 13 000 | 8 000        | 12 000 | 10 000    | 13 000 | 10 000  | 14 000 | 8 500          | 11 500 | 10 000         | 15 000 | 8 500          | 13 000 |
| Service Engineer      | 6 000         | 9 000  | 5 500             | 8 500  | 5 500  | 8 500  | 6 500  | 9 000  | 5 500  | 8 500  | 8 000         | 11 000 | 8 000   | 12 000 | 5 500  | 8 500  | 5 500        | 8 500  | 5 500     | 9 000  | 6 500   | 10 000 | 6 200          | 8 800  | 6 500          | 11 000 | 5 500          | 8 000  |
| IT                    |               |        |                   |        |        |        |        |        |        |        |               |        |         |        |        |        |              |        |           |        |         |        |                |        |                |        |                |        |
| IT Manager            | 16 000        | 22 000 | 14 500            | 23 000 | 13 000 | 22 000 | 15 000 | 22 000 | 14 000 | 22 000 | 16 000        | 25 500 | 16 000  | 26 000 | 15 000 | 22 000 | 13 000       | 19 000 | 14 000    | 22 000 | 16 000  | 23 000 | 13 000         | 19 000 | 16 000         | 25 000 | 14 000         | 23 000 |
| IT Specialist         | 6 500         | 9 500  | 6 500             | 9 000  | 6 500  | 8 000  | 7 000  | 9 500  | 6 000  | 8 500  | 8 000         | 11 000 | 8 000   | 11 000 | 6 000  | 9 500  | 6 000        | 8 000  | 6 000     | 9 500  | 7 500   | 10 000 | 6 500          | 8 500  | 7 500          | 10 000 | 7 000          | 9 500  |
| ERP Specialist        | 7 000         | 10 000 | 7 000             | 10 000 | 6 500  | 8 500  | 7 000  | 10 000 | 6 500  | 9 500  | 8 000         | 11 000 | 8 500   | 12 000 | 7 000  | 10 000 | 6 500        | 8 500  | 7 000     | 10 000 | 8 000   | 10 000 | 6 500          | 9 000  | 8 000          | 11 000 | 8 000          | 10 000 |
| Network Administrator | 7 000         | 11 000 | 7 000             | 11 000 | 7 000  | 9 500  | 8 000  | 11 000 | 7 500  | 10 000 | 8 000         | 12 000 | 9 000   | 12 000 | 6 500  | 11 000 | 6 500        | 9 000  | 7 000     | 11 000 | 8 000   | 10 000 | 6 500          | 9 000  | 8 000          | 11 000 | 8 000          | 11 000 |
| System Administrator  | 7 000         | 12 000 | 7 000             | 11 000 | 7 000  | 9 500  | 8 000  | 11 000 | 6 000  | 9 000  | 8 000         | 12 000 | 9 000   | 12 000 | 6 500  | 11 000 | 6 500        | 9 000  | 7 000     | 11 000 | 8 000   | 10 000 | 6 500          | 9 000  | 9 000          | 11 000 | 8 000          | 11 000 |
| Developer             | 10 000        | 15 000 | 9 000             | 14 000 | 9 000  | 13 000 | 10 000 | 14 000 | 9 000  | 13 000 | 11 000        | 16 000 | 11 000  | 16 000 | 9 000  | 14 000 | 9 000        | 13 000 | 9 000     | 15 000 | 10 500  | 15 000 | 10 000         | 13 000 | 10 000         | 15 000 | 10 000         | 14 000 |



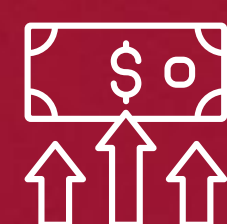
Salary report – industrial sector 2022

## Key findings



12.1%

of houses is vacant, according to preliminary results of the National Census



+11.4%

average salary increase in Automotive and Appliances group among researched provinces



+7.6%

population increase after the influx of Ukrainian immigrants, according to Unia Metropolii Polskich Association



+10.5%

average salary increase in R&D group among researched provinces



+9.6%

average salary increase in the Back Office group among researched provinces



+13%

average salary increase in Chemical & FMCG group among researched provinces



A close-up photograph of a person's hands holding a rugged, dark-colored handheld device. The device has a screen displaying a grid of data and a keypad with blue and white buttons. A red emergency stop button is visible on the top. The person is wearing a silver watch on their left wrist. The background is a blurred industrial setting with metal structures and machinery.

## Challenges faced by the Manufacturing Sector





# Katarzyna Kołuda

Division Manager

## Candidate Experience – recruitment challenges in the manufacturing industry

What is the candidate experience and why is it so important in the era of shortage of specialists with technical competences? It's a very wide outlook on the experiences of both employees and candidates - from the moment of publishing an offer, through subsequent stages of candidate selection and employment. The recruitment process should not be treated as something marginal or extraneous, as oftentimes it forms a source of opinions on the given employer. In the era of common Internet access, social media portals, communicators and industry forums, who among us hasn't been asked to provide an opinion on a product or service? Opinions on employers are just as common as those about holiday destinations or home appliances to buy. When seeking a new job or participating in the recruitment process, we oftentimes look for such opinions among friends and in the Internet. The opinions we find in the web can have a notable impact on one's opinion on a given company and the final decision of whether to agree to work in it.

On the current job market, such opinion is created by the employer - that is the director, the manager, the recruiter, the employee. Each of these people should be aware of their impact on success of the recruitment process. Up to 3/4 of physical workers and almost as many specialists have declared willingness to share their opinion on the recruitment process they took part in. The opinion begins forming when candidate reads the recruitment offer. An increasing number of candidates is paying attention to whether the offer is readable, contains reliable information on the scope of obligations





and requirements and if the information on salary, work system and employment terms is transparent. According to research published in the eRecruiter report, up to 67% of respondents have answered that they want the employers to publish pay range in their job offers. Sending a candidate's resume is only a first step to success. That is why it's so important for the candidate to know, already at the application stage, that their resume did not fall into a void. The candidates want to know whether they made it to the next recruitment stage or what has caused them to be rejected. It might seem obvious, yet the reality of recruitment is oftentimes unable to match these basic expectations. Meanwhile, the atmosphere of a race for the talent, shortage of required competences on the labor market and unwillingness to chance jobs due to perceived unstable situation in Poland and in the broader world are being increasingly palpable. Should a candidate decide to change their job, especially in case of a professional aware of their market position, they will most probably apply to several, perhaps over a dozen, employers. It should be little surprise that with several offers to choose from, they want to receive feedback and pass the recruitment process as quick as possible. Already a quarter of candidates want to know the reasons of their rejection immediately.

Once a candidate is invited to partake in the recruitment process, we should take care to plan the process out appropriately. Time is of the essence here. In case of recruiting mid-level employees and specialists, the optimal timeframe of the recruitment process, from submitting the application to the final employment decision, should not exceed two weeks. In case of mid-level management or C level positions, the recruitment process usually requires several stages, additional tests and tasks and so the candidates expect it might take longer. However, we should not extend it indefinitely. We should try to gather several decisive people to the same meetings to limit the number of necessary stages and the time pressure on candidate. The assumption that candidate's ability to attend several interviews at different times indicates their motivation is outdated. We all value our time and meticulously fill our schedules trying to maintain work-life balance. In case of physical workers, the most effective recruitment process involves: recruitment decision taken during a single meeting, direct contact with the recruiter, ability to get familiar with the workplace and one's direct manager. Physical workers will appreciate a physical meeting over a virtual one, which might prove more challenging and stressful due to lack of appropriate tools or computer skills.

This trend looks somewhat different in case of specialist and management employees. This group definitely prefers remote meetings for their initial interviews. As for the second recruitment stage, physical meetings are usually favored.

Regardless of the stage, one should take care to maintain a good, positive and consistent communication with the candidate. This concerns both clearly presenting the course of the recruitment process to the candidate during an early stage, the expected number of stages and their timeframe. We should state what to expect during the interview – if it involves additional tests or tasks, we should provide their outline. We also shouldn't omit the information about other people present in the interview. If the meeting will take place in a company's office, one should provide any relevant tips about the place and directions. The less stressful a job interview is for the candidate, the more willing they'll become to talk freely and honestly.

Let's return to the issue of feedback information, as it is important in each stage of the recruitment. The physical workers usually prefer quick, specific feedback. Specialist and managers, on the other hand, believe they should get more information on the reasons for their rejection. Based on research conducted by the eRecruiter portal, over 84% of respondents recruited for specialist positions appreciate it when feedback includes competence and experience assessments. 54% of respondents would like to know about the subjective feelings of the recruitment manager and their assessment of matching the organization and its values.

The last, but certainly not least, aspect of candidate experience is the onboarding process, which builds a candidate's knowledge of the organization, introduces them to the company, its values, daily tasks and rituals. A reputation for well-planned onboarding has become an increasingly important argument for the candidates to accept an offer. This reduces their fears related to adapting to the new work environment. Furthermore, many employers start building their relations with new employees even before they come to office, rather than waiting with onboarding for their first day of work.

In the area of candidate experience, many employers utilize surveys to gauge satisfaction of candidates during the recruitment process. It allows for researching weak and strong aspects of the process and implementing changes. It's also worthwhile to take care of the career tab on the company website. Aside from current job offers, it should contain information about available benefits, opportunities for growth, EB activities, values, employee events, employee opinions about the company and their everyday tasks and growth. A candidate entering such a website should be able to learn as much as possible about the company from the perspective of its employees.

To sum up, the candidate experience challenges to be faced by employers in the near future include:

- describing best practices for people conducting recruitments and having implemented standards of conduct,
- trainings, workshops and other regular internal activities within the scope of candidate experience, conducted primarily among the management staff,
- maintaining access to information,
- a professional and transparent approach to meetings with candidates,
- consistent communication and providing constructive feedback on each stage of recruitment,
- researching candidate opinions,
- well-planned and coordinated onboarding,
- strong Employer Branding and building positive reputation among employees and candidates.

*\* Based on data from the eRecruiter.pl - Candidate Experience Report in Poland 2021.*





**production**  
manager

# Przemysław Ozga

Editor in Chief

## Challenges faced by industry in 2022

The third decade of the 21st century has stirred us from the dream of globalization's unshakeable paradigm. The last thirty years have been unprecedented on the global scale. We lived in a sanctuary, in which global trade and supply chains have become natural elements of the business landscape. Product individualization or building competitiveness while maintaining high quality and margins have spurred decisions towards globalization of production. The consequences of these decisions are now shaking the economies and budgets of many countries. We have lived a western lifestyle, while utilizing Asian labor. The dependence on Taiwanese semiconductors, Indian pharmaceutical components or Russian resources was a natural outcome of developed service economies and Western companies wishing to utilize cheap labor in their race for a better tomorrow.

The pandemic and undermining of US dominance by Russia and China (a child of the globalization) have initiated the process of deglobalization, more precisely: fragmentation of global regions in at least several areas:

- industry (possible return of industry to the West, increased production agility),
- economy (new pathways and diversification of resource flow),
- currency (attempts to undermine US dolar, recurring interest in cryptocurrencies),
- military (alliances opposed to NATO).





It is of interest to Poland's industrial sector, which thanks to the initial momentum of the economy at large has noted a growth of over 17 percent following the invasion of Ukraine (as of March 2022) despite high inflation. However, the indicators for new orders in April have caused some concern, as they had shown the quickest fall since November 2020. Companies have informed that month of high cost inflation, at some of the highest levels on record. The prices of most means of production have risen, especially those of natural resources, metals and fuels. Furthermore, many basic packaging processes underwent nearshoring and were „delegated” to our eastern neighbor - Ukraine, Belarus and Russia, which are currently embroiled in a war. All of this creates price instability and reduces convergence between supply and demand.

The reduced dependence of Polish industry on the automotive sector, whose bottlenecking has been impacting economies of states such as Germany - Europe's largest producer - since the outbreak of the pandemic, is a positive signal. Utilizing the opportunity of a large influx of female Ukrainians is a great challenge ahead of Poland. Over 56% of women fleeing the war has higher education - it is a great potential requiring a structural approach from both Polish manufacturing companies and the state itself (teaching language, support for adaptation on the job market, the risk of excessive long-term social help). Polish economy, including the industrial, logistics and trade sectors could immediately adapt over 700,000 employees. Since 2014, Ukrainians have been responsible for 12-13 percent of Poland's GDP growth, forming one of the driving forces for the development of Polish business.

The time of a potential deglobalization and return of some industries to Europe are, in my opinion, a chance for Poland's economy. By using the technological goods of the digital transformation, modern education, Ukrainian emigration and possible production relocation from Asia, the country could avoid the middle-income trap and smartly harness the ongoing crisis.





# Radostaw Drzewiecki

CEO

## Leadership in 2022

We live in a time when the effectiveness of leaders is being constantly tested. The costs of living are rising and companies are struggling with a shortage of qualified personnel, disrupted supply chains and, as of late, uncontrolled cost increases. Moreover, people are voluntarily leaving their jobs, even with no other position lined up. This trend, called the Great Attrition will only get worse, which is why so much rests on true leaders. It is them who currently constitute one of the key links in the process of organization's growth and who create an effective, engaging workplace that greatly impacts the performance of teams and companies at large, regardless of the industry.

The key challenge faced by leaders in 2022 is combining empathy and adaptability. The empathy understood as genuine interest in people, ability to look from their perspective and providing the right conditions for them to feel as a part of the organization. In a research conducted by McKinsey on a sample of 4,924 workers (from Australia, Canada, Singapore, Great Britain and United State of America), 40% of the respondents declared plans to leave their job within 3-6 months. The reasons to do so included lack of appreciation (54%), the direct manager (52%) and lack of sense of belonging (51%). Nowadays, we compete not only for the clients, but also for the employees. Employees choose workplaces with leaders who know how to build team identity to support broader company identity. This is why an empathetic leadership is so important. Leadership at every level, especially at the most neglected line manager level, should be distinctive and deliberate. This lets the employees decide whether they want to choose whether they want to be a part of a given team within the company.



## McKinsey Quaterly, September 2021

Aside from empathy, adaptability constitutes another key factor. It is closely related to the management, that is a leader's day-to-day work. The empathy is closely linked with leadership understood as the ability to create a sense of belonging to the team, linking people with company's broader strategy, its business environment. Therefore leadership should be deliberate and management effective. A modern leader is unafraid of what's happening on the market and instead harnesses the crises to obtain a competitive advantage, both in the context of competing for the clients and for the employees.

Compensating for cost increases and avoiding loss of profitability through increased process effectiveness is a true challenge nowadays. This task primarily requires the leaders to understand their companies on a business level. The adaptability begins with gaining familiarity with processes, their measurement and observation, in order to act on facts rather than opinions. It is a tedious work and, regrettably, companies often opt not to do it. Yet, according to our audit, a properly conducted diagnosis constitutes 90% of a successful transformation. It enables a company to learn its potential for increasing effectiveness and implement changes based on solid information. However, to call a leader's adaptability above-average, they should be constantly questioning the status quo, focus on productivity and perpetual improvement. Such an approach provides a genuine competitive advantage.

Today, more than ever, we need strong leaders who understand that empathy is a strength, rather than a weakness. We need leaders who can lead deliberately and manage effectively, building an above-average ability to adapt.





# Łukasz Górecki

Director of the SA&AM Cluster,  
Katowice SSE Centre for Strategic Investments

## The Fourth Industrial Revolution

The Fourth Industrial Revolution is, without a doubt, one of the key challenges faced by Polish businesses. Despite dealing with it for almost a decade, in Poland we still identify:

- low digitization levels of companies, low interest and readiness to implement Industry 4.0 technologies,
- growing need to meet price and quality challenges, which cannot be reached by modern industrial businesses without the use of advanced automation, robotization and digitization of production systems,
- focus of Industry 4.0 technology providers on working with large companies conducting complex, labor-intensive processes and lower investment readiness of SMEs.

These factors cause a large number of SMEs - a key part of our economy - to modernize at an insufficient pace. If this situation will not change, Polish businesses might lose their competitiveness in the coming years, which we cannot allow. It is crucial to swiftly implement mechanisms of systemic support for industrial transformation of the SME sector. One of the tools of such mechanisms are Digital Innovation Hubs - regional centers supporting digital transformation, which are already being created in Poland. In accordance with European Union's digitization policy, Digital Innovation Hubs are to become strong, neutral partners in the digital transformation processes, especially in the case



of SMEs. Thanks to these hubs, SMEs will be able to learn new technologies and their use in the company based on specific case studies and see how they're able to increase work effectiveness and the quality of produced goods. Digital Innovation Hubs are able to provide consultancy, train employees and support the entire implementation process.

In February 2022, European Commission has closed recruitment for the contest to create Digital Innovation Hubs in member states. The winners will receive the status of an European Digital Innovation Hub (EDIH). Aside from the prestige and ability to utilize the EDIH network, the selected centers will receive financial support to facilitate their services. In Poland, 25 candidates have entered the contest, of which up to 16 can receive the award. One of them is the Silesia-based Silesia Smart

System, coordinated by Katowice SSE. Silesia Smart System is already offering a wide selection of support services for SMEs, including:

- providing demonstrators to facilitate simulations, tests and verifying potential solution before their implementation,
- auditing processes, business models and goals of the digital transformation, preparing digital transformation strategy,
- consulting and intermediation in the scope of financial engineering,
- training services concerning sharing information and best practices concerning the digital transformation and implementation of its highly specific process, organized jointly with technology providers and integrators,
- intermediacy in making contacts and working with technology providers and integrators within the scope of designing, testing and implementing the new solutions.

The digital transformation is no longer a choice, but rather a necessity. Many Polish companies (especially smaller ones) in a stable situation and with secured contracts does not see a need for the transformation and does not treat it as a priority. It might impact their competitiveness and future

market position. Today, the Silesia Smart System DIH, coordinated by Katowice SSE is a strong and neutral partner for small and medium-sized Silesian companies in their digital transformation process. Through awareness-raising activities, demonstrations, trainings, consultancy and intermediacy in contacts with the suppliers, it provides full support for the digital transformation process. This allows small and medium-sized businesses from our region feel catered for. They gain a chance to increase their effectiveness and maintain competitiveness on European and global market.





# Krzysztof Warzecha

President of the Board

## Industry 4.0. in Poland – a subjective perspective

There are businesses in Poland (primarily the large ones) on high level of implementation of Industry 4.0 technologies. There are also some businesses (primarily small ones or conducting processes of low repeatability or with a large involvement of manual operations) still about to reach the second industry revolution. In a global scale, is this good or bad? I'd like to leave you with the impression it's not bad at all.

According to Deloitte, one Polish company in five belongs to the digital vanguard. The extent to which digital technologies have been implemented differ not only between the businesses, but also between areas. The most advanced implementations appear to belong to the area of maintenance, where predictive techniques are common. Having said that, the next level - prescriptive, allowing for fully forecasting the costs - is rarely seen.

While automation and robotization of processes can be implemented in an evolutionary way, operation by operation and process by process, building a digitized business oftentimes requires changes to the basics - which might be quite revolutionary for a given company. Creation of a data lake and a structure for analytical applications with direct access to source data requires abandoning process thinking, which has been a bedrock of the four basic levels of automation used up to this point. It also requires a shift towards a multi-vector IoT structure - a costly process involving risk to some of the senior management that is difficult to assess.





Polish businesses are implementing innovations based on source technologies, developed abroad. When I was invited to the Smart Remote Services conference in Berlin two years ago, I was the only speaker from outside DACH countries. Polish companies utilize platforms developed by American, German or French tech companies and rarely create their own, similarly advanced platforms. This is despite the basics of artificial intelligence, as it is used currently, have been known in Poland for decades. Polish market is still quite shallow in the area of digitization, for several reasons.

I believe that there are five key reasons slowing down the development of Industry 4.0 in our country. The first of these is still common lack of systematic knowledge of digitization. The second is focus on short-term profits many businesses have. The third - low presence of professional management in family-owned SMEs, whose founders refuse to cede control despite reaching retirement age. The fourth reason lies in lacking ability to budget ROIs for digitized processes. The fifth and final, critical and long-term reason is lack of personnel and a training system for Knowledge Intensive Industries, which in the coming years will prove a key factor bottlenecking the growth of Poland's economy and making it more difficult to shed the reputation as Germany's cheap production base.

Additional difficulty comes from the fact that digitization is a somewhat undefined term - its shape and meaning is shaped fully by a company's individual digital transformation strategy. After all, it will look completely different for a small company implementing RPA to leave behind Excel (or even a paper notebook) and for a business introducing DT of a full technological cycle to accelerate and lower the costs of future R&D processes. As an aside, both these approaches can cause a major headache when trying to estimate potential returns. After all, what are the chances that the creativity of a clerk freed from Excel will impact company's ROE? Will the inherently high-risk R&D activities propel the company to success, freeing it from the need to deftly maneuver in a race for the lowest margins? The difference between the speed of developing technologies and the capacity to grow the company definitely requires VUCA-style management.

To conclude, I'd like to note that industry development will not stop at the fourth level. In future perspective, Industry 5.0 is still ahead of us - a stable, human-centric and environmentally friendly vision of manufacturing systems charted by the European Commission. I heartily recommend reading EC's report to better understand what lies in the future.





Ewa Latkowska

Country HR Director Poland

### **Manufacturing Industry in 2022 – limitations and opportunities**

Business risks and challenges ahead of entrepreneurs in 2022 have already been described in detail by economists and market analysts. Regrettably, the obstacles brought on by the pandemic (such as component shortages and supply chain disruptions) haven't been solved yet while new difficulties arose: price and salary pressure, dramatic increase in prices of electricity, natural gas and fuel. An additional challenge comes from the uncertainty of conducting business when a war rages past our eastern borders. Subsequent embargoes on goods and resources produced by Russia and changing geopolitical relations will bring additional challenges to manufacturing companies not only in Poland or Europe, but across the world.

However, the pandemic has also brought new opportunities to manufacturing companies, by accelerating their digital transformation, which is surely key for effective and flexible business management based on reliable data and effective analytics. In an uncertain micro- and macroeconomic environment, given a variable availability of components, resources, their prices and production costs, only reacting quickly to impending changes can ensure success of businesses. Reliable data and monitoring of the processes is also a good way to create savings, which might prove crucial for retaining business profitability given the rising production costs.





The digital transformation is also a chance to develop the company and its employees. Taking the company I work in as an example, I can confidently say that Industry 4.0, 3D printing, robotization and automation - both in the manufacturing and warehousing space form a highly enticing development alternative for the employees. On the other hand, the digital revolution also brings new challenges in the need to upskill and reskill the employees. Businesses currently undergoing the digital transformation are already struggling with a lack of specialists on the market. Therefore, another great challenge to occupy the manufacturing industry throughout the near future is to train their employees, develop their competences and help them gain new skills.

When talking about training, one ought to mention educating the youth. Based on many years of experience, I can safely say that the changes to Poland's education system do not guarantee a generational change in manufacturing companies, especially in the areas of specialized production and machine operation. Only a small fraction of students is interested in learning technical jobs. The once closed technical and vocational secondary schools are being created anew, but this is a mere drop in the ocean of needs of employers implementing new technologies. In our company, we have invested in a proprietary education system to suit our own purposes. To that end, we have organized the Flex School, in which we train new technicians and specialists, teaching them vocational skills in the environment of a manufacturing facility. We also cooperate with over a dozen technical schools and universities in Pomerania. This cooperation encompasses, among others, regular meetings, workshops or financing stipends for the best students. We can see that such action contributes to a flow of qualified candidates on a shrinking job market.

Another challenge currently faced by employers is the intergenerational cooperation. In many companies, a single department or team might include people from four different generations: baby boomers about to end their professional careers, generation X, Y and Z. How can one satisfy their differing expectations and approaches to work? Is the unwillingness to work overtime a sign of laziness or a conscious choice arising from one's outlook on life and the opinion that while work might be important, it is not more important than one's family or personal accomplishment? The businesses

and their bosses should learn and understand the needs of various candidates and manage them in a deliberate way, avoiding the trap of judging and grading various ways of thinking or differing expectations.

In a time of uncertainty and economic risk, effective leadership is paramount. This is a challenge that the employers currently face and will keep facing in the coming years. Effective leaders should be flexible and adjust their management style to the developmental needs of their employees. They should be able to effectively convey instructions and specific information, but also act as coaches and support the growth of their teams. In light of dynamic transformations, including the digital transformation, the ability to lead people through changes is also key. Empathy has become critical, as it allows us to look from differing perspectives without judging, as well as to listen carefully and support the team through an unstable time. In Flex, we have been investing in training the leaders to communicate and work with empathy. We build a space for reflection, sharing experiences and learning from each other. We train key leadership skills based on real challenges faced by our leaders. I believe that a strong, diverse team that trusts each other and has the knowledge and competences fit for its current and future needs will guarantee a company's successful operation and growth on an unpredictable market. Much like the effective leadership, adjusted to the modern challenges.





# TOYOTA

## Marcin Szczepblewski

HR Manager

### // The manufacturing industry is full of challenges on its own

The manufacturing industry is full of challenges on its own - modern manufacturing systems require careful planning, adhering to numerous norms, procedures and rules, as well as providing appropriately high quality. This already makes recruitment in this area difficult and, is further complicated by its oftentimes mass character, highly competitive job market need to appropriately train the newly hired teams and motivate them to avoid negative outcomes such as employee turnover or absences.

For over two years we have been struggling with the coronavirus pandemic. While for many industries it was a pretext to fully utilize the opportunities provided by remote and hybrid work, the ability to implement such solutions is limited for the manufacturing industry. The employers have to strive (and bear costs) to reduce the risk of infection spreading while also dealing with growing impatience of the crews tired by the limitations and restrictions. The regulations concerning action against the pandemic are often of an administrative and static character (such as the number of tests and registered infections), with the real consequences of infections and subsequent problems being borne by employers and employees.

The speed with which COVID-19 has spread across the world is not the only side-effect of globalization - supply chain disruptions are another price we pay for living in the „global village”. The blocked Suez Canal, fires in North America and difficulties with component production in Asia all contributed to





product unavailability (an inconvenience for customers) and fluctuations of production. The latter are forcing the companies to flexibly react to changing demand and production capabilities, adjusting the form and volume of employment.

The last „shock” on the job market has been caused by the war in Ukraine, which - as of writing - is highly uncertain as to how and when will it end. We can also see an impact on all aspects of company's operation: business (difficulties related to the flow of goods and services to and from Ukraine), social (outflow of employees leaving to fight and an influx of refugees) or image building (working with Russian partners or operating on Russian market).

Employers have to face not only these global upheavals, but also local challenges of Poland's job market. The country's economic situation might seem relatively stable for an average citizen, which appears to be reflected by the public media. However, the noticeable price increase due to inflation creates expectations of increased remuneration, which - in addition to the above mentioned external factors - will create internal pressure on the employers.

Who, then, will be able to prosper in such fickle, unpredictable circumstances? The companies which have been managing their internal communications in a responsible way over the years, creating an understanding of the goals, challenges and directions will definitely have an advantage. It is also important to build deliberate leadership on all levels of management and maintain a transparent company policy, especially in the areas of employment and remuneration.



# Adam Pustelnik

First Deputy-Mayor of the City of Łódź

## // Łódź has made a strong bet on developing the manufacturing industry.

An example of that is the urban planning project letting the city offer approx. 300 ha of new lots within the city's administrative borders, in direct proximity to the airport and the almost-finished S-14 route completing the highway ring around Łódź. These lots are owned by developers, entities offering the opportunity to work in accordance with the highest standards and providing both space for rent and under the model of ultimately transferring the ownership of developed property to the investor. In the recent months, we have seen yet another disruption to the supply chain, which in case of Łódź is oftentimes reliant on rail transport running across areas that are currently under armed warfare of economic sanctions. In the short-term, this is an adverse situation, as it disrupts the rhythm of supplies, disorganizing operations of companies. However, in the long term, all these changes might result in a permanent transformation of the logistic web of interdependencies – its regionalization in place of globalization, or in other words a return to the formula that worked for decades. We see a great opportunity for Łódź in this, as it is a perfect place not only for manufacturing certain goods, but also their distribution across the entire Europe.

Łódź not only boasts the best transport connections in all of Poland, but also offers the highest level of regional public support among all of Poland's cities.





When presenting the city's profile in the context of industrial investments, one has to mention the maturity of the local market. Historically, we have always been an industrial city and have experienced an influx of foreign investments and development of local businesses for a long time. Our market is characterized by a certain diversification: we are a national production leader in textile, chemical and home appliances segments. Łódź is often described as risky from a demographical standpoint, but in the recent years it had seen approx. 100,000 new people taking residence here, which is rarely acknowledged by methodologies focused on official numbers of registered residents.

To the industrial investors, we are offering a chance to work with and draw on the potential of Łódź University of Technology - one of the foremost technical universities in Poland. It is of key importance in shaping the cadres that, in future, will fill the ranks of white collar workers. On the other hand, the relatively high unemployment rate hovering around 5% proves the scale of possibilities available to investors seeking employees for less demanding roles.

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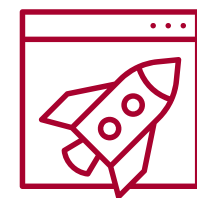
**9**  
Branches across  
the country

**122**  
Employees within  
our structures



**50000**

Active Engineering  
and Production engineers  
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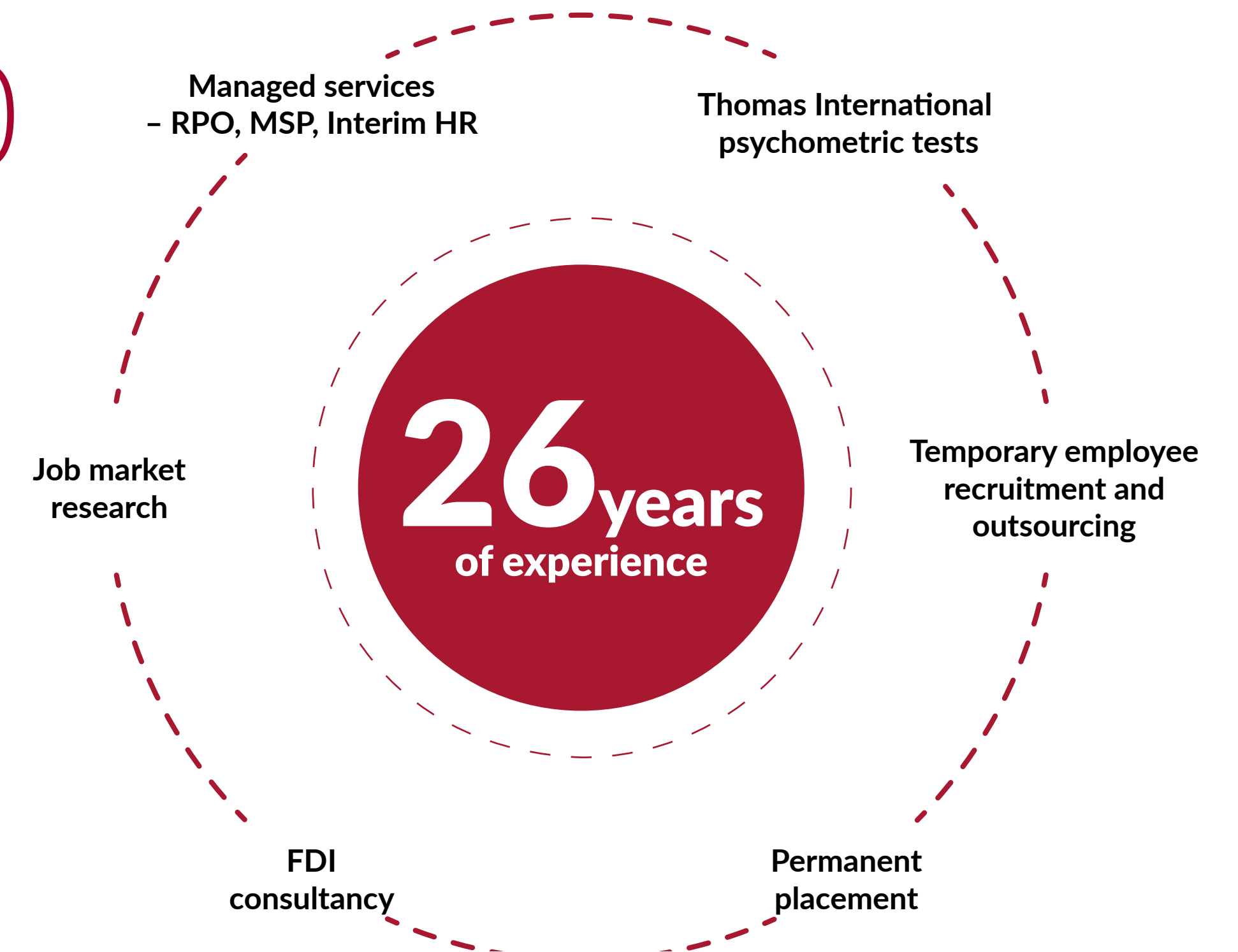
**66**

Completed HR projects  
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**73**

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We're at your disposal – get in touch with us.

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